Geometric Programming for Aerodynamically-Actuated Wingsail Design Optimization

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***Abstract*—In this article, we describe a deterministic optimization framework for the conceptual-stage design of aerodynamically-actuated wingsails. The primary objective of high-performance sailing is well understood: maximize the conversion of unsteady aerodynamic forces into forward thrust, without inducing excessive overturning moments. However, designing a sail to meet this goal is by no means straightforward due to the existence of multiple recursive, nonlinear design relationships. Consequently, most wingsails are designed in an iterative fashion, using some combination of linear heuristics and engineering intuition. This approach can produce viable designs, but it does so at the expense of time and capital, and provides little physical insight into the underlying design space. By formulating the wingsail design problem as a geometric program, it is possible to quickly generate hundreds of optimal candidate designs, assess their sensitivity to specific constraints and parameters, and determine the shape of Pareto frontiers. Unlike general nonlinear optimization methods, geometric programming optimization is computationally efficient, and requires no initial guesses or hyperparameter tuning. Perhaps most importantly, all the decision variables in a geometric program are determined simultaneously, eliminating the need for iterative piecewise optimization of subsystems. These benefits come at a price: all objective and constraint functions must be described as posynomials. Nevertheless, we demonstrate that this restricted set of functional forms can adequately capture the key physical relationships between wingsail parameters, and provide quantifiable physics-based design guidance.**

***Index Terms*—Geometric programming, optimization, sail, sailing, wing, wingsail.**

NOMENCLATURE

$()_W$	Wing variable.
$()_T$	Tail variable.
$()_B$	Nose ballast variable.
A	Aspect ratio.
b	Wingspan [m].
$\bar{c}_{\text{MAC}}$	Mean aerodynamic chord [m].
$\bar{c}$	Mean geometric chord [m].
c_r	Root chord [m].
c_t	Tip chord [m].
c^*	Optimal chord distribution [m].
$\tilde{c}$	Nondimensional chord distribution [m].
C_m	Pitching moment coefficient.
C_{foil}	Foil profile wetted perimeter [m].
C_{L_α}	Lift-slope constant [1/rad].
$\mathcal{C}$	GP cost function.
$\mathcal{C}_\lambda$	Taper ratio cost function.
d_T	Aerodynamic center separation [m].
C_D	Total 3-D drag coefficient.
C_d	Profile drag coefficient.
C_{D_i}	Induced drag coefficient.
C_{D_f}	Skin friction drag coefficient.
C_l	2-D lift coefficient.
C_L	3-D lift coefficient.
D	Aerodynamic drag force [N].
e	Oswald efficiency factor.
E	Taper optimization error function.
g	Gravitational acceleration [m/s^2].
H	Vessel side (heeling) force [N].
J	Mass moment of inertia about mast axis [kg $\cdot$ m^2].
k	Aerodynamic spring constant [1/N $\cdot$ m].
k_1	b/z_r.
k_2	z_r/z_0.
K	Total roll moment [N $\cdot$ m].
K_a	Aerodynamic roll moment [N $\cdot$ m].
K_g	Gravitational roll moment [N $\cdot$ m].
K_h	Hull stability roll moment [N $\cdot$ m].
l_b	Vessel beam width [m].
l	Sectional lift force [N].
l^*	Optimal sectional lift force [N].
L	Aerodynamic lift force [N].
m	Mass [kg].
N	Total yaw moment [N $\cdot$ m].
N_{CW}	Clockwise yaw moment [N $\cdot$ m].
N_{CCW}	Counterclockwise yaw moment [N $\cdot$ m].
p	$1 + 2\lambda$.
R	Aerodynamic resultant force [N].
Re	Reynolds number.
S	Wing planform area [m^2].
S_{foil}	Airfoil profile area [m^2].

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S_{wet} Wing surface area [m^2].
$\bar{t}$ Mean airfoil profile thickness [m].
$t_{\max}$ Max airfoil profile thickness [m].
T Vessel thrust force [N].
$\boldsymbol{u}$ Vector of all decision variables.
U_{10} Wind speed, 10m reference [m/s].
U Wind speed, center of effort [m/s].
u^* Atmospheric boundary layer friction velocity [m/s].
$V\!\!\!/$ Wing volume [m^3].
x_{AC} Distance between mast axis and aerodynamic center [m].
z_0 Atmospheric boundary layer roughness length [m].
z_{CE} Elevation of aerodynamic center [m].
z_{CG} Elevation of center of gravity [m].
z_r Elevation of wing root [m].
z_t Elevation of wing tip [m].
α Angle of attack [rad].
δ Deflection angle [rad].
η Nondimensional spanwise coordinate.
Γ Airfoil circulation [m^2/s].
κ_A Airfoil area constant.
λ Taper ratio.
λ^* Optimal taper ratio.
ν_1 $(\lambda^2+\lambda+1)/(\lambda+1)^2$.
ν_2 $(3\lambda^2+2\lambda+1)/(\lambda^2+\lambda+1)$.
ν_3 $(2\lambda+1)/(\lambda+1)$.
ϕ Roll angle [rad].
ρ Air density [kg/m^3].
ρ_c Wing core volumetric density [kg/m^3].
ρ_s Wing skin surface density [kg/m^2].
τ Airfoil thickness ratio.

I. Introduction

The past 20 years have witnessed a marked increase in the use of autonomous surface vessels (ASVs) for oceanographic research and naval defense purposes [1], [2]. However, the range and endurance of most ASVs remains limited by finite onboard energy reserves, most of which are consumed in the process of overcoming hydrodynamic drag. A growing number of uncrewed platforms are able to harness kinetic energy directly from the wind for the purposes of locomotion—or, simply put, to *sail*. This dramatically extends the duration and reach of their deployments.

It is important to note that autonomous sailboats are no longer an academic curiosity—they are a vital component of many ocean observing networks [3]. Existing commercial platforms have already traversed the Pacific and circumnavigated Antarctica, proving their potential value for oceanographic research [4]. However, much of their potential remains untapped. Indeed, the notable diversity of existing architectures evinces the relative immaturity of this nascent field of robotics, and hints at the existence of unrealized efficiency and performance gains. One sail variety that is frequently implemented on uncrewed platforms is the free-rotating, aerodynamically-actuated wingsail.

The first aerodynamically-actuated wingsail was developed by German aviation engineer Flettner [5] in 1926. This design involves a pair of rigidly connected, vertically oriented lifting surfaces, which are free to rotate about a fixed shaft embedded within the upwind wing. The downwind wing, often referred to as the *stabilizer*, *tail*, or—as Flettner called it—the *auxiliary plane*, stabilizes the upwind wing by generating a restoring torque in response to flow disturbances. The basic architecture of an aerodynamically-actuated wingsail is shown in Fig. 1, along with the aerodynamic forces and moments they generate. When both wing and tail chord lines are parallel, the system aligns itself with the mean flow field; however, when the tail is deflected, it generates an aerodynamic torque that drives the leading wing to settle into a new equilibrium about a nonzero angle of attack. Once there, the wing is able to produce useful lift. In this way, an aerodynamically-actuated wingsail closely resembles the longitudinal pitch control system of an aircraft.

This arrangement has two major advantages over mechanically-actuated wingsails: open-loop stability and energy efficiency. As was alluded to in the preceding paragraph, aerodynamically-actuated wingsails automatically maintain a fixed angle of attack relative to the apparent wind, eliminating the need for high-bandwidth closed-loop control schemes, and the costly sensors and actuators they require to function properly. This characteristic "weather vane" behavior allows the wingsail to gently respond to changes in wind direction, preventing accidental gybes which can capsize or destroy a vessel. Aerodynamically-actuated wingsail systems tend to be remarkably well-damped, and can be designed to achieve desired setpoint tracking and disturbance rejection properties [6]. They also consume very little power. Instead of using electromechanical actuators to wrestle the wingsail away from its equilibrium orientation, aerodynamically-actuated wingsails leverage fluid forces to achieve an advantageous equilibrium. However, these admirable characteristics are underwritten by a delicate balance of unsteady, nonlinear aerodynamic forces, and are by no means guaranteed for all aerodynamically-actuated wingsail architectures. The performance of a given system is determined by dozens or hundreds of interdependent design parameters, making rigorous physics-based design challenging.

Aerodynamically-actuated wingsails have become increasingly popular on both commercial and academic uncrewed sailing platforms [7]. A recent review of uncrewed sailing platforms conducted by An and Yu [8] concluded that the vast majority of these wingsails are designed using a combination of sea trial results and engineering experience, but lack strong theoretical underpinning. This *trial-and-error* approach can be effective; however, it is also time-consuming, expensive, and provides little insight into the underlying design space. Nevertheless, virtually every wingsail in existence was designed in this manner: a Reynolds number, airfoil profile, and aspect ratio are assumed a priori, with little or no justification, and used to estimate coefficients of lift and drag numerically; these coefficients are then used to determine a planform area, which—it is hoped—will generate a desired thrust force at a specified wind speed [9], [10], [11], [12], [13], [14], [15]. Inconveniently, Reynolds number is itself a function of planform geometry, and the aerodynamic performance of a given airfoil profile is highly dependent on the Reynolds number regime in which it operates. This tangled web of recursive design relationships explains why wingsail

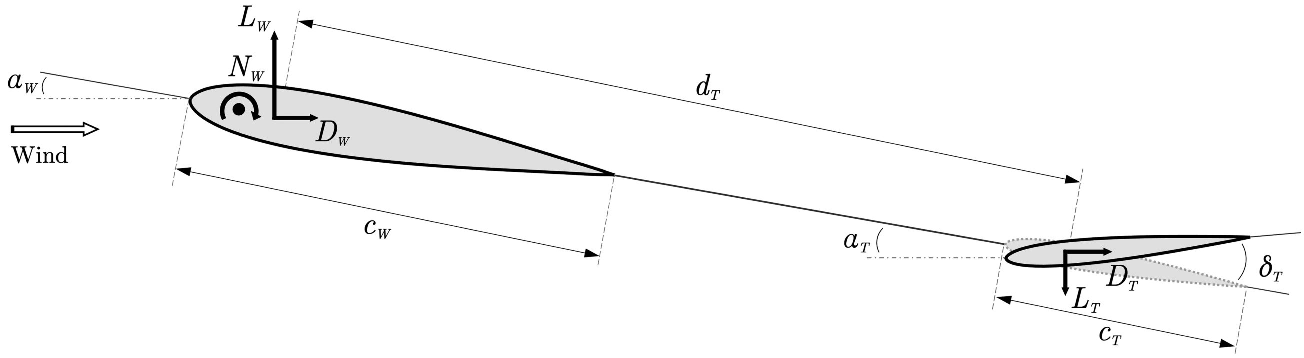

Fig. 1. Top down view of aerodynamic forces and moments produced by an aerodynamically-actuated wingsail at equilibrium.

designers have historically relied upon apocryphal heuristics for guidance.

The proposed methodology attempts to minimize the number of a priori assumptions involved in wingsail design, and in turn, generate better designs and more realistic estimates of performance. This goal is accomplished using an optimization technique called *geometric programming*. The solution of a geometric program (GP) determines the value of each decision variable simultaneously, making it particularly well-suited to handle multidisciplinary design optimization (MDO) problems involving highly coupled parameters and recursive design relationships. Modern GP solvers leverage primal-dual interior point methods, which provide local sensitivity information at no additional cost; this feature is useful, given that intuition often fails in high-dimensional parameter space. Perhaps most importantly, the solution to a GP is guaranteed to be globally optimal with respect to the chosen objective function and constraints. Formulating the wingsail design problem as a GP eliminates the need for iterative velocity prediction program computations, and enables the rapid discovery of wingsail architectures that maximize upwind performance without exceeding a specified roll moment. This manuscript details the first comprehensive, physics-based wingsail design optimization methodology that obviates the need for preliminary assumption of Reynolds number, airfoil profile, and aspect ratio. The proposed methodology also accounts for the atmospheric boundary layer, which is especially important to consider when designing smaller wingsail systems.

The rest of this article is organized as follows. The basic structure of a GP is described in Section II, and the primary constraints pertaining specifically to wingsail optimization are presented in Section III. In Section IV, the optimal taper ratio for a wing subject to a logarithmic spanwise velocity profile is determined for the first time, and subsequently formulated as a GP-compliant constraint. The proposed GP is solved using *GPkit* [16], a Python package for defining and manipulating geometric programming models. The resulting optimal wingsail design is provided in Table I. For most MDO problems, a single optimal solution is not sufficient; rather, we seek to understand the value of certain tradeoffs, and the sensitivity of the objective function to particular decision variables. Accordingly, a set of Pareto frontiers and sensitivity analyses are presented in Section V, and discussed further in Section VI alongside limitations of the proposed approach. Finally, Section VII concludes this article.

II. Geometric Programming

Optimization often requires a balance between model generality and solver efficiency. On one end of the spectrum, linear programs (LPs) are extremely computationally efficient and produce globally optimal solutions, but impose severe limitations on the types of functions that can be used to formulate the problem; on the other, there are nonconvex, iterative routines, such as genetic, surrogate, and particle swarm optimization, which can solve arbitrarily complex design problems, but are computationally expensive and provide no guarantee of global optimality.

GPs lie somewhere in between these two extremes. In fact, GPs are generalized LPs, and can be solved nearly as quickly: sparse GPs with tens of thousands of decision variables, and millions of constraints, can be solved in a few minutes on a desktop computer [17]. GP optimization is also compatible with a wide variety of nonlinear relationships that arise often in various fields of engineering, and has been leveraged to optimize the design of electronic circuits [18], communication systems [19], aircraft [20], and mass-transit networks [21]. Finally, and perhaps most importantly, the solution to a GP is guaranteed to be globally optimal.

A GP is a type of constrained nonlinear, nonconvex optimization problem, in which the objective can be described as a posynomial function, and the constraints can be described as either monomial equalities or posynomial inequalities. In 1967, Duffin et al. [22] discovered that GPs enjoy a peculiar property: they become convex when subject to a logarithmic change of variables. Indeed, log transformation converts monomials into affine functions, and posynomials into a sum of exponentials of affine functions, both of which are convex; accordingly, any optimization problem that can be formulated as a GP can be solved using modern convex optimization techniques, such as primal-dual interior point methods [23]. Further, unlike general nonlinear optimization approaches, GP optimization requires no initial guesses or hyperparameter tuning, and produces solutions that are globally optimal with respect to the specified objective function and constraints. A GP in standard form has the following structure:

$$\begin{aligned} \text{minimize:} \quad & f_0(\boldsymbol{u}) \\ \text{subject to:} \quad & f_i(\boldsymbol{u}) \leq 1, \quad i = 1, \ldots, p \\ & h_i(\boldsymbol{u}) = 1, \quad i = 1, \ldots, m. \end{aligned} \tag{1}$$

The term $f_0(\boldsymbol{u})$ in (1) is the *posynomial cost function*, and represents a quantity to be minimized, subject to a set of *posynomial inequality constraints*, $f_i(\boldsymbol{u})$, and *monomial equality constraints*, $h_i(\boldsymbol{u})$, which are described in terms of a vector of *decision variables*, $\boldsymbol{u}$. Together, these components define a GP. The term posynomial is a portmanteau of the words "positive" and "polynomial," and describes a function composed of the sum of an arbitrary number of single-term monomials;[1] a monomial, in turn, is simply the product of an arbitrary number of strictly-positive variables raised to real-valued powers, multiplied by a strictly positive coefficient

$$h(u) = cu_1^{a_1}u_2^{a_2}\dots u_n^{a_n} \qquad \{c \in \mathbb{R}_{++},\ a_j \in \mathbb{R}\}. \tag{2}$$

Equation (2) can be written in a more compact form using power law notation

$$h(\boldsymbol{u}) = c\boldsymbol{u}^{\boldsymbol{a}} \tag{3}$$

$$\boldsymbol{u}^{\boldsymbol{a}} = \prod_{j=1}^{n} u_j^{a_j}. \tag{4}$$

Using this notation, a posynomial takes the form

$$f(\boldsymbol{u}) = \sum_{k=1}^{K} c_k \boldsymbol{u}^{\boldsymbol{a}_k} \qquad \{c_k \in \mathbb{R}_{++},\ \boldsymbol{a}_k \in \mathbb{R}^n\}. \tag{5}$$

The exponents a_j in (2) can be any real numbers—fractional, negative, or otherwise—but the coefficient c must be positive. A few examples of monomials include: 0.707, $2u_1$, $\sqrt{u_1/u_2}$, and $3u_1^2u_2^{-0.83}$. As noted in [20], the equation for lift, $L = 0.5\rho U^2 S C_L$, is a monomial in $\boldsymbol{u} = [\rho, U, S, C_L]$, with $c = 0.5$ and $\boldsymbol{a} = [1, 2, 1, 1]$. Various relationships pertinent to aerospace engineering are described by monomial functions, including lift-induced drag, skin-friction drag, and Reynolds number.

Posynomials offer a versatile means of modeling various real-world phenomena, allowing for the representation of complex relationships among decision variables. Posynomials are closed under addition, multiplication, and nonnegative scaling, while monomials are closed under multiplication and division [23]. When a posynomial is multiplied by a monomial, the product remains a posynomial; likewise, dividing a posynomial by a monomial yields another posynomial. These properties are extremely useful, as they permit a broad range of extensions to the standard GP formulation given by (1). For example, the constraint $f(\boldsymbol{u}) \leq h(\boldsymbol{u})$ can be rewritten as $f(\boldsymbol{u})/h(\boldsymbol{u}) \leq 1$, while the constraint $h_1(\boldsymbol{u}) = h_2(\boldsymbol{u})$ can be rewritten as $h_1(\boldsymbol{u})/h_2(\boldsymbol{u}) = 1$. Various relationships pertinent to aerospace engineering can be described by generalized posynomial functions, including wing volume, surface area, and mean aerodynamic chord.

A. GP Modeling

If an optimization problem can be expressed as a GP, it can be solved quickly and easily; furthermore, the solution is guaranteed to be globally optimal with respect to the objective function and constraints. However, the primary challenge becomes describing the objective function and constraints in a GP-compliant fashion, or so-called *GP modeling*. While many physical relationships are well described by posynomial inequalities and monomial equalities, many still are not. Fortunately, GP-incompatible functions can often be massaged into a GP-compliant form by algebraic manipulation or a substitution of variables, while others can be approximated by surrogate functions, fit to discrete data using nonlinear least squares regression [24].

In standard form, GPs only support monomial equality constraints; this can be problematic, given the preponderance of posynomial (and polynomial) equality constraints in engineering design problems. For instance, as noted in [20], aerodynamic drag is often decomposed into a profile drag component and an induced drag component, resulting in a posynomial equality constraint

$$C_D = C_d + \frac{C_L^2}{\pi A e} \tag{6}$$

where C_d is the 2-D profile drag coefficient, C_L is the 3-D lift coefficient, A is the aspect ratio, and $e \in [0:1]$ is the Oswald efficiency factor, which accounts for the additional drag incurred by wings with non-elliptical spanwise lift distributions. This expression is not GP-compatible; however, by leveraging knowledge of the GP structure, it can be handled using a technique referred to as *posynomial equality relaxation*, and expressed in a GP-compliant manner

$$C_D \geq C_d + \frac{C_L^2}{\pi A e}. \tag{7}$$

Posynomial equality relaxation transforms a strict posynomial equality into GP-compliant posynomial inequality based on the premise that, under specific conditions, the optimal solution will still satisfy the original equality. The first condition requires that the decision variable—C_D in the example above—is not already constrained by other monomial equalities that would prevent the optimization procedure from reducing, or *tightening*, its value until the original equality condition is satisfied. Of course, this tightening will not occur unless the cost function is applying downward pressure on this variable. Posynomial relaxation requires that the objective function, and all other posynomial inequalities, are monotone decreasing in the decision variable in question. For the purposes of the example given above, drag frequently appears in the cost function, providing the requisite downward pressure. Essentially, from an intuitive perspective, before attempting to perform posynomial equality relaxation, one must ask, "Is the cost function applying downward pressure on the variable in question, and is there anything preventing its reduction, other than the relaxed posynomial inequality constraint?" If the cost function is applying downward pressure, and there are no obstructions to tightening, the original equality constraint will be satisfied.

It is important to acknowledge not all functional forms are amenable to these techniques. For example, phenomena described by trigonometric functions and polynomial equalities

[1] Note, however, that a posynomial is not simply a positive polynomial; while the latter are restricted to positive integer powers, the former can be raised to any real-valued power.

involving negative terms are infamously challenging to model using GP-compatible functions. Problems involving such relationships can be formulated as signomial programs (SPs), and solved in an iterative manner, but global optimality is no longer guaranteed [25]. Prior to resorting to describing the problem as an SP, it is preferable to either simplify the problem, or approximate the relationship of interest using GP-compliant functional forms.

III. GP Wingsail Design

Before embarking upon a wingsail design optimization effort (or any optimization effort for that matter) it is necessary to first contemplate the true objectives, and assess the degree to which these goals can be encoded in mathematical language. The objective (or cost) function is the backbone of the GP, and drives every decision variable to its constrained optimum value. Seemingly analogous objectives can yield surprisingly disparate outcomes. For instance, in sailing, one might seek to maximize the ratio of thrust-to-heel force; maximize total lift; or minimize sail mass—each of these objectives will require different constraints and approximations, and will produce markedly different solutions.

A. Cost Function

The choice of objective function depends not only on the true goal, but also upon the generality of the optimization methodology employed. For instance, if formulating the wingsail design problem as an unconstrained optimization problem, one might be inclined to attempt to maximize the ratio of thrust-to-heel force; this is an admirable approach, and yields fascinating results [26]. However, our true goal is better represented by a constrained optimization problem, in which we seek to maximize thrust, subject to a heel force constraint. After all, the true goal in sailing is not to minimize heel force, but rather to ensure it does not exceed the vessel's finite capacity to counter it at a representative roll angle. This objective was well articulated by Schererer [27], who in 1974 stated that "For upwind [sailing], it is clearly desirable that the sail should develop the highest possible thrust, up to wind speeds where roll stability or increased hull drag become important." The performance of a wind-powered vessel is often benchmarked by its ability to make headway upwind; downwind sailing, while useful and often necessary, is objectively easier to accomplish due to a more advantageous alignment of aerodynamic forces. Thus, it is appropriate to seek a wingsail design which facilitates thrust generation while sailing upwind. This is particularly true if the wingsail is to be mounted on an uncrewed platform which cannot benefit by the fast decision-making and real-time guidance of a skilled human operator. Vessel thrust is defined by the expression

$$T = L \sin(\beta) - D \cos(\beta) \tag{8}$$

where L is the wingsail lift, D is the wingsail drag, and β is the angular difference between the vessel and the apparent wind, often referred to as the *apparent wind angle* or *apparent point of sail*. Inconveniently, this expression includes both trigonometric functions and a negative coefficient, and thus

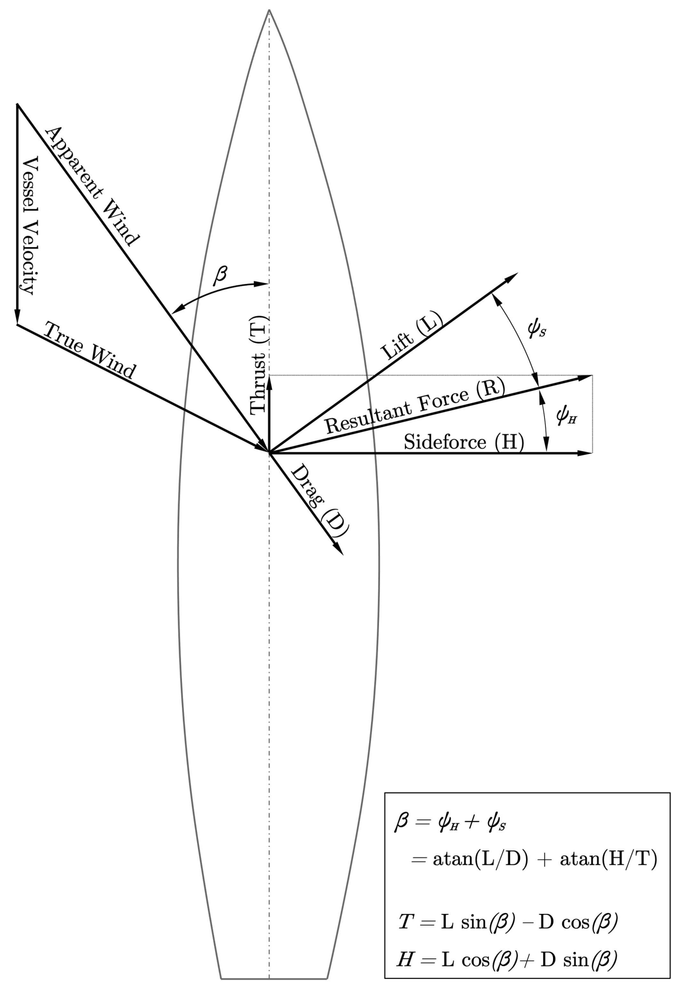

Fig. 2. Decomposition of aerodynamic forces into thrust and sideforce. Adapted from [27].

cannot be used as a GP objective function; consequently, it is necessary to formulate a surrogate function that captures the spirit of this objective. Inspection of (8) reveals that for upwind pointing angles ($\beta < \pi/2$), thrust monotonically increases with lift, and monotonically decreases with drag.[2] This supposition is confirmed graphically by Fig. 2, which reveals that a larger *lift-over-drag ratio* creates a more favorable alignment between the aerodynamic resultant force (R) and the thrust axis of the vessel. Importantly, this ratio—which is often used as a proxy for aerodynamic efficiency—can be expressed in GP-compatible posynomial form, making it an attractive surrogate function. It will be shown in Section V that the maximization of lift-over-drag is not exactly analogous to upwind thrust maximization; however, this metric adequately captures the spirit of the true objective, and yields physically meaningful design recommendations.

[2]When sailing just upwind of a beam reach ($\beta \approx \pi/2 - \epsilon$), an increase in lift will generate far more thrust than a commensurate reduction in drag. It is nevertheless beneficial to minimize drag at all upwind pointing angles in order to satisfy roll stability constraints.

The lift produced by the tail of an aerodynamically-actuated wingsail (L_T) does not contribute significantly to vessel thrust at steady state; however, the drag incurred by the tail (D_T) can be substantial, and must be considered. The *total lift-over-drag ratio* is defined here as the lift produced by the wing (L_W), divided by the drag incurred by both the wing and tail ($D_W + D_T$), and its inverse is the posynomial cost function ($\mathcal{C}$) employed in the proposed GP architecture

$$\boxed{\mathcal{C} = \frac{D_W + D_T}{L_W}}. \tag{9}$$

Lift and drag are formulated in the same manner for both the leading wing and the tail

$$L = \frac{1}{2}\rho U^2 S C_L \tag{10}$$

$$D = \frac{1}{2}\rho U^2 S C_D \tag{11}$$

and the substitution of (10) and (11) into (9) yields

$$\mathcal{C} = \frac{C_{D_W}}{C_{L_W}} + \left(\frac{S_T}{S_W}\right)\frac{C_{D_T}}{C_{L_W}}. \tag{12}$$

When the cost function is expressed in this form, it becomes apparent that the optimization procedure will, in the absence of any constraints, attempt to drive the wing planform area to infinity, while reducing the tail planform area to zero.[3] Of course, this is not particularly helpful design guidance, and underscores the necessity of physically meaningful constraints. The size of the wing is limited by the overturning moment it creates, while the size and position of the tail must be determined in accordance with the aerodynamic moments produced by the wing, such that the full wingsail system is able to quickly and precisely react to changes in wind direction.

B. Decision Variables

Prior to encoding the wingsail design problem as a GP, it is wise to first determine a minimal set of decision variables, and express all constraints in terms of this parameter set. This increases interpretability and computational efficiency, as it reduces the dimension of the optimization domain. A single-tapered[4] wing can be described using many parameters, including: planform area (S), wingspan[5] (b), mean geometric chord ($\bar{c}$), mean aerodynamic chord (c_{MAC}), root chord (c_r), tip chord (c_t), mean thickness ($\bar{t}$), maximum thickness ($t_{\max}$), vertical center of mass (z_{CG}), and vertical center of effort (z_{CE}). These parameters are all illustrative, but many of them are redundant. A generic symmetric (NACA00xx) wing with no sweep, twist, or spanwise variation in airfoil profile can be fully defined using only four variables: planform area (S), aspect ratio (A), taper ratio (λ), and thickness ratio (τ). The definition of each parameter is provided in (13)–(16), and the variables that comprise them are displayed graphically in Figs. 3 and 6

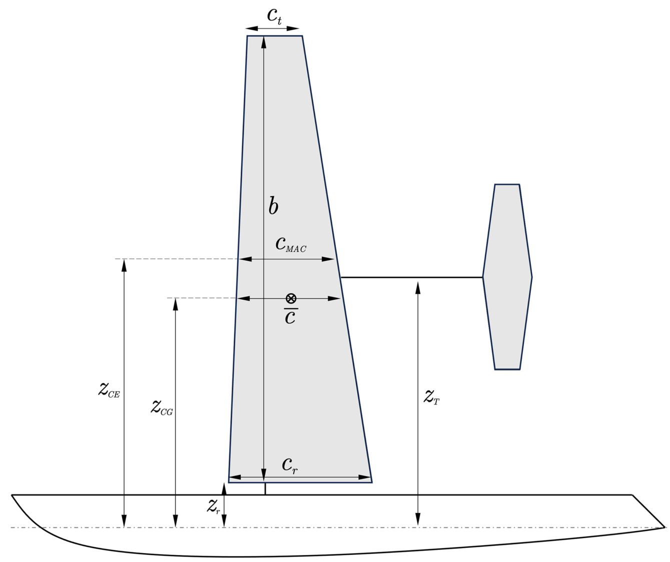

Fig. 3. Aerodynamically-actuated wingsail planform dimensions. All variables shown are defined for both the wing and tail.

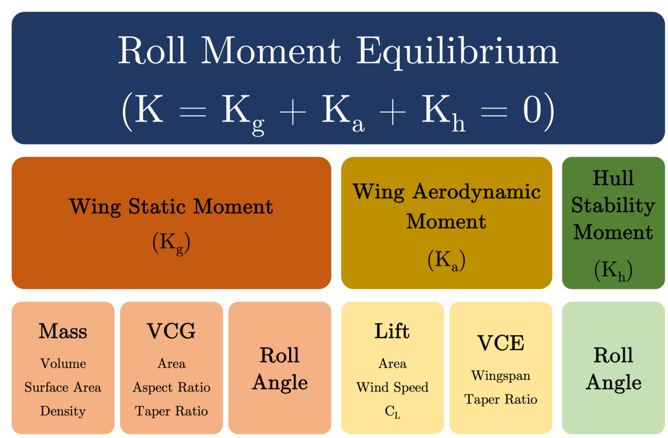

Fig. 4. Hierarchical expansion of the roll equilibrium constraint.

$$S = b\bar{c} \tag{13}$$

$$A = b^2/S \tag{14}$$

$$\lambda = c_r/c_t \tag{15}$$

$$\tau(z) = t_{\max}(z)/c(z) \qquad z \in [z_r, z_r + b]. \tag{16}$$

Planform area is the projected area enclosed by the wing's leading edge, trailing edge, root chord, and tip chord; aspect ratio is the square of the span divided by the planform area (or, equivalently, the span divided by the mean geometric chord); the taper ratio is the length of the tip chord divided by the length of the root chord; and the thickness ratio is the maximum airfoil

[3]In the absence of constraints, this cost function also permits a solution involving a small wing and a *very* small tail; however, this outcome is precluded by the physical relationship between lift, drag, and Reynolds number. This is relationship is discussed further in Section III, and illustrated in Fig. 8.

[4]A single-tapered wing is one in which the chord length is reduced linearly from root to tip; conversely, a multitapered wing exhibits a piecewise linear reduction of chord from root to tip.

[5]In the context of wingsail design, the term *wingspan* is used to indicate the height of the wing, from root to tip; in aerodynamics texts, this is often referred to as the *half-span*.

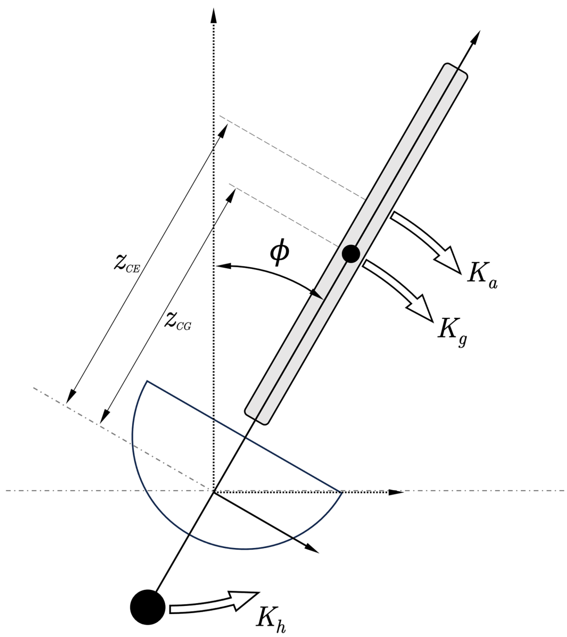

Fig. 5. Illustration of the moments contributing to the wingsail roll equilibrium: K_h is the stabilizing moment provided by the hull and keel; K_g is the static gravitational moment produced by the weight of the wingsail, applied at the wingsail center of gravity (z_{CG}); and K_a is the aerodynamic torque generated by the flow of air around the wingsail, applied at the wingsail center of effort (z_{CE}).

profile thickness at a given spanwise station, divided by the chord length at that station. The thickness ratio is a fixed property of an airfoil profile, and thus it will be constant for any wing with a uniform airfoil profile. For symmetric NACA00xx sections, the point of maximum thickness is located at the 30% chord, measured from the leading edge.

All other decision variables related to a generic wing can be described in terms of this minimal set. Thus, an aerodynamically-actuated wingsail can be fully defined by two sets of these parameters, corresponding to the leading wing and tail, in addition to the tail separation distance (d_T).

C. Selected Models

Constraints perform two roles in a GP: first, they allow the designer to explicitly set limits on certain decision variables, reflecting a priori knowledge of the system. For example, it is possible to constrain the height and separation of an inchoate candidate tail of unknown dimension, such that its lowermost corner will not intersect the flotation plane at a given yaw and roll angle. This is the traditional use of the term "constraint." However, constraints can also be used to encode the key physical relationships in a GP, and inform the model how each of the decision variables relate to one another. The most important models are described in the following subsections, and additional models are provided in Appendix A.

1) Roll Equilibrium: The fundamental constraint involved in the proposed wingsail GP formulation ensures roll stability by limiting the overturning moment induced by the wingsail. The optimal balance between sailing speed and roll stability has not been determined [8]; however, it is generally understood that the goal of sailing is to harness as much kinetic energy from the wind as possible, without overwhelming the righting moment provided by the hull and keel. Accordingly, the roll equilibrium constraint for the proposed GP was designed to guarantee universal roll stability under a possible *worst-case scenario:* a poorly executed tack maneuver. In this situation, the vessel is pointing directly into the wind while the wingsail is generating lift at peak efficiency, such that the lift vector and the sideforce vector are completely aligned. In this configuration, the full wing area—not a projection thereof—is generating lift, and thus there is no spanwise flow or *spilling* of wind off the wing with increasing roll angle. Further, it is assumed that the vessel has been heeled over to ninety degrees by wind, waves, inertia or some combination thereof, such that the full mass of the wing is contributing to the overturning moment. If the vessel is able to self-right under these conditions, it will be stable in any other configuration at the design wind speed, provided the deck is sealed and the wingsail is positively buoyant.

Roll equilibrium is defined by the sum of three components, which can be rearranged to form an inequality constraint

$$K_h \geq K_g + K_a \tag{17}$$

where K_g is the wingsail static gravitational moment, K_a is the wingsail aerodynamic moment, and K_h is the hull stability moment. In order to preserve roll stability, the gravitational and aerodynamic moments must not exceed the hull stability moment at steady state. The hierarchical composition of the roll equilibrium constraint is displayed diagrammatically in Fig. 4, and graphically in Fig. 5. Each of these relationships shall now be analyzed in turn.

The static moment produced by the weight of the wingsail is given by

$$K_g \geq g \sin(\pi/2) \sum_{i \in \{W,T\}} z_{CG_i} m_i \tag{18}$$

where the index $i \in \{W, T\}$ is used to designate the wing and tail components. The static overturning moment produced by a given wingsail is determined by its mass and center of gravity elevation. The mass of a wing is a function of its surface density, wetted surface area,[6] internal core density, and volume; volume is, in turn, a function of taper ratio, aspect ratio, thickness ratio, and planform area, as well as the shape of the chosen airfoil profile; finally, the wetted surface area can be described in terms of the wing planform area and thickness ratio. In order to assess the gravitational overturning moment, models describing these basic geometrical properties of a wingsail must first be derived.

2) Airfoil Profile Area: For any given airfoil profile described by chord length (c) and thickness variable (y), the 2-D profile area is given by

$$S_{\text{foil}} = \int_0^c \left[y_{\text{upper}}(x) - y_{\text{lower}}(x)\right] dx. \tag{19}$$

[6]In naval architecture, the *wetted surface area* is the portion of a vessel hull that is immersed in water; however, in this and other aerospace applications the term is used to describe an area which is in contact with the external airflow.

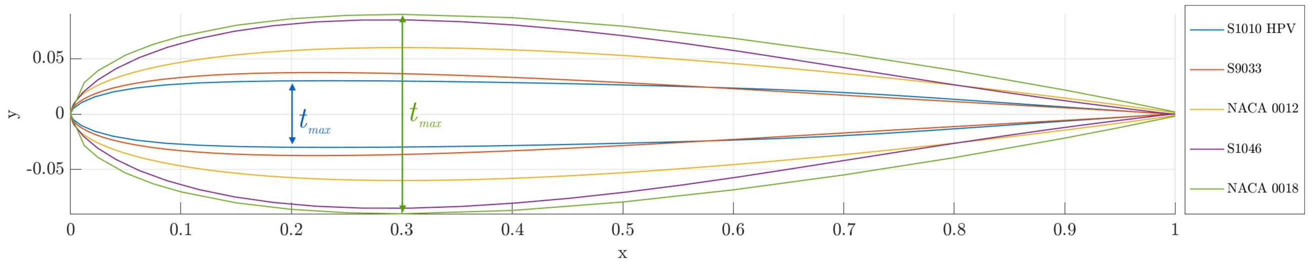

Fig. 6. Assorted symmetric airfoil profiles with thickness ratios varying from 0.06 to 0.18. The chord length of each section is normalized, and the maximum thickness ($t_{\max}$) is shown for the thickest and thinnest airfoil sections.

The airfoil thickness ratio is defined by $\tau = t_{\max}/c$, where $t_{\max} = \max\{y_{\text{upper}}(x) - y_{\text{lower}}(x)\}$, as illustrated in Fig. 6.

The foregoing relationships suggest that S_{foil} likely varies in proportion to c and $t_{\max}$, scaled by some constant κ_A to be determined [28]. This observation motivates the formulation of a simple proportionality relationship

$$S_{\text{foil}} \sim \kappa_A c t_{\max} = \kappa_A c^2 \tau. \tag{20}$$

Let us set the simplified airfoil profile area (20), equal to the true area (19), in order to determine the proportionality constant, κ_A

$$\kappa_A c^2 \tau = \int_0^c \left[y_{\text{upper}}(x) - y_{\text{lower}}(x)\right] dx$$

$$\kappa_A = \frac{1}{c^2\tau} \int_0^c \left[y_{\text{upper}}(x) - y_{\text{lower}}(x)\right] dx. \tag{21}$$

All symmetric NACA00xx series airfoils are parameterized by a closed-form equation

$$y = 5\tau[0.2969\sqrt{x} - 0.1260x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4] \tag{22}$$

where $x \in [0 : 1]$ is the independent chordwise variable, and y is the profile half-thickness at x. By substituting (22) into (21) and solving the integral, the profile area proportionality constant is determined

$$\kappa_A = 0.685. \tag{23}$$

The accuracy of this linear approximation varies from airfoil to airfoil, but is generally quite acceptable. When applied to symmetric NACA00xx series airfoils, the linearized area model incurs less than 0.7% error. There is no significant categorical difference in error between symmetric and cambered foils, nor is there a strong correlation between thickness ratio and accuracy.

3) Wing Volume: Having established an acceptable linearized expression for the cross-sectional area of a given airfoil profile, it is now possible to derive an expression for wing volume by computing a spanwise integral of the airfoil profile area. Assuming a uniform wing (i.e., no spanwise profile variation), κ_A and τ can be removed from the integral, yielding an expression for wing volume

$$\forall = \int_0^b S_{\text{foil}}(z)dz = \kappa_A \tau \int_0^b c(z)^2 dz. \tag{24}$$

For a single-taper wing, the chord at any given spanwise position $z \in [0, b]$ can be described by the expression

$$c(z) = c_r - \left(\frac{c_r - c_t}{b}\right) z \tag{25}$$

where c_r is wing root chord, c_t is the wing tip chord, and b is the wingspan. This relationship is interpretable but redundant, and can be described using fewer parameters by expressing the spanwise coordinate z in nondimensional form, $\eta = z/b \in [0 : 1]$, and recalling the definition of the taper ratio ($\lambda = c_t/c_r$). The nondimensional formulation of the chord equation in (25) is

$$c(\eta) = c_r + c_r\eta(\lambda - 1). \tag{26}$$

In order to compute the volume integral using the nondimensional chord expression (26), it is necessary to reflect the change of variables using the chain rule, which gives $dz = b \times d\eta$. Therefore, the volume integral, in terms of η, is given by

$$\forall = \kappa_A \tau \int_0^1 c(\eta)^2 b d\eta. \tag{27}$$

Solving this integral gives an expression for wing volume

$$\forall = \frac{4\kappa_A S^{3/2}\tau}{3\sqrt{A}} \nu_1(\lambda) \tag{28}$$

$$\nu_1(\lambda) \geq \frac{\lambda^2 + \lambda + 1}{(\lambda + 1)^2} \tag{29}$$

where S is the planform area, A is the aspect ratio, λ is the taper ratio, and τ is the thickness ratio. Note that ν_1 is not a GP-compliant posynomial; however, it is well-approximated by a fitted two-term softmax affine function, as shown in Fig. 7. This relationship can then be relaxed to form a valid generalized posynomial inequality constraint.

4) Wing Surface Area: The wing surface area model is a simplified adaptation of the expressions used in aerospace design and engineering textbooks [29], which are obtained by spanwise integration of the continuously varying *wetted perimeter length* of each airfoil section. The wetted perimeter length is defined here as the sum of the lengths of the top and bottom splines of an airfoil profile (i.e., $y_{\text{upper}} + y_{\text{lower}}$). A reasonable approximation of the wetted perimeter length of most subsonic sections is given by

$$C_{\text{foil}} \approx 2c(1 + 0.25\tau). \tag{30}$$

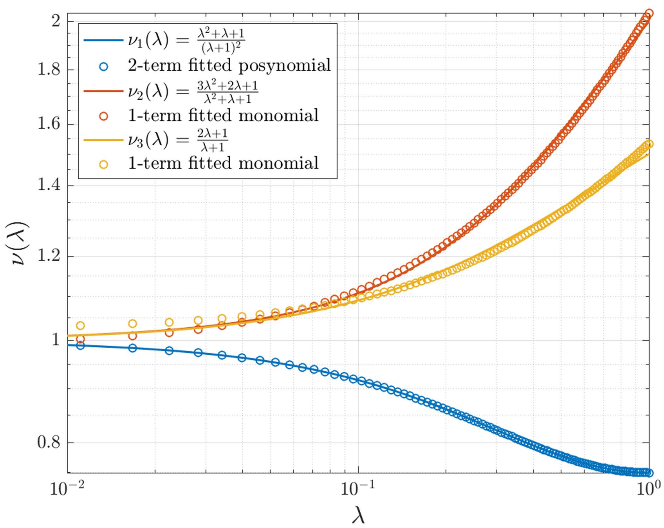

Fig. 7. Geometric relationships related to wing volume (ν_1), vertical center of gravity (ν_2), and vertical center of effort (ν_3), as a function of taper ratio. These functions are not inherently GP-compliant, but are well approximated by posynomials. The root mean squared percentage error of each fitted function is 0.0024% (ν_1), 0.051% (ν_2), and 0.051% (ν_3).

When used to approximate the wetted perimeter length of a NACA0018 airfoil profile with $c = 1$, this approximation incurs a proportional error of only 2.3% (compared to direct numerical summation of panel lengths). The wetted area of the wing is then computed by spanwise integration of the wetted perimeter function (30)

$$S_{\text{wet}} \approx 2S\left[1 + 0.25\tau_r \frac{1 + \frac{\tau_t}{\tau_r}\lambda}{1+\lambda}\right] \tag{31}$$

where τ_r and τ_t correspond to the thickness ratio at the root and tip, respectively. For a wing with uniform thickness ratio (i.e., $\tau_r = \tau_t$), (31) can be simplified further, and takes the form

$$\begin{aligned} S_{\text{wet}} &\geq 2S + b\left(\frac{t_{\max}}{2}\right) \\ &\geq 2S\left(\frac{\tau}{4} + 1\right). \end{aligned} \tag{32}$$

This equation effectively models a wing as a thin isosceles triangular prism with wetted area equal to twice the planform area, plus half the projected frontal area.

5) Wing Mass: The wing mass model assumes that the wing is homogeneous, with known core density (ρ_c) and skin density (ρ_s) defined according to user-specific structural robustness requirements and manufacturing capabilities. These two parameters are used in conjunction with the wing volume and surface area models derived in the previous two subsections to estimate total wing mass

$$m \geq \rho_c \forall + \rho_s S_{\text{wet}}. \tag{33}$$

6) Vertical Center of Gravity: The final model which must be determined in order to estimate the wing's gravitational overturning moment is the vertical center of gravity (z_{CG}); the vertical centroid of a solid object of uniform density is given by

$$\bar{z} = \frac{1}{\forall}\int_{\forall} z\, d\forall. \tag{34}$$

Accordingly, the centroid of a wing can be approximated by an integral equation of the form

$$\begin{aligned} z_{CG} &= z_r + \frac{1}{\forall}\int_0^b z S_{\text{foil}}(z)\,dz \\ &= z_r + \frac{1}{\forall}\int_0^b z[\kappa_A c(z)^2 \tau]\,dz \\ &= z_r + \frac{1}{\forall}\int_0^1 (b\eta)[\kappa_A c(\eta)^2 \tau] b\,d\eta. \end{aligned} \tag{35}$$

Evaluating the integral in (35) gives the following relaxed posynomial equality relationships that describe the wing vertical center of gravity, evaluated relative to the plane of flotation

$$z_{CG} \geq z_r + \frac{\sqrt{SA}}{4}\nu_2 \tag{36}$$

$$\nu_2 \geq \frac{3\lambda^2 + 2\lambda + 1}{\lambda^2 + \lambda + 1}. \tag{37}$$

The nonposynomial function ν_2 is shown in Fig. 7, alongside its monomial surrogate function.

Having characterized the gravitational overturning moment using models describing the basic geometrical relationships involved in wingsail design, it is now necessary to analyze the aerodynamic forces involved in the roll moment equation. The following aerodynamic models are multipurpose, and are also used to define the objective function.

7) Reynolds Number: The Reynolds number is the foremost nondimensional quantity used to define broad classes (or *regimes*) of fluid motion. The magnitude of the lift and drag forces generated by a given airfoil are dependent on the chord Reynolds number; thus, for the purposes of wingsail design, it is important to know how the Reynolds number varies in response to alterations in wing planform geometry. This relationship is described by a monomial equality

$$Re = \frac{\rho U}{\mu}\sqrt{\frac{S}{A}}. \tag{38}$$

8) Lift and Drag: Lift is quite amenable to mathematical description: it is mostly linear at low angles of attack, and is not particularly sensitive to Reynolds number; further, lift is only slightly affected by three-dimensional effects like downwash and wingtip vortices.[7] Accordingly, in the proposed GP architecture, it is assumed that the 2-D profile lift coefficient (C_l) is equal to the 3-D lift coefficient (C_L).

The same cannot be said of drag: indeed, drag varies nonlinearly with angle of attack, and is highly dependent on Reynolds number, especially in the low-Reynolds number regime ($Re < 1e6$). Accordingly, higher fidelity semi-empirical drag

[7] In practice, the downwash behind finite-span wings induces a local reduction of the angle of attack, resulting in a decrease in the overall lift coefficient [30]. However, this effect is small and often ignored when operating in low-Reynolds number flows.

models are often employed, which decompose drag into three components: *profile drag*, *skin-friction drag*, and *lift-induced drag*. Profile drag (C_d), also known as *form drag* or *pressure drag*, is an inviscid phenomenon that arises purely from the pressure differential between the front and the back of an object as it moves through a fluid. Skin-friction drag (C_{D_f}), or *viscous drag*, is caused by friction between an object's surface and the fluid it moves through. Finally, lift-induced drag (C_{D_i}), or simply *induced drag*, is a 3-D effect, caused by the pressure field discontinuity at the tips of finite-span wings. This discontinuity gives rise to wingtip vortices, which reduce both the effective angle of attack and effective span of a wing. Each of these components can be parameterized using a combination of empirical and theoretical relationships and summed to create a total drag model, which can then be applied to both the wing and tail in the proposed GP architecture

$$C_D \geq C_d + C_{D_i} + C_{D_f} \tag{39}$$

$$C_d \geq C_d(C_L, Re, \tau) \tag{40}$$

$$C_{D_i} = \frac{C_L^2}{\pi A e} \tag{41}$$

$$C_{D_f} = \frac{S_{\text{wet}}}{S}\left(\frac{0.074}{Re^{1/5}}\right). \tag{42}$$

The profile drag coefficient (C_d) is not known analytically, and thus must be determined empirically, or estimated using numerical approaches. To this end, aerodynamic coefficients were computed offline using *XFOIL*, a viscous panel method solver for 2-D airfoil analysis [31]. All simulations were carried out assuming incompressible flow (i.e., Mach number equal to zero), and used 320 panels to characterize the airfoil pressure distribution. For each of 15 symmetric NACA00xx airfoils with thickness ratio (τ) ranging from 0.08 to 0.22, the profile lift, drag, and moment coefficients were computed at 33 angles of attack, $\alpha \in [0 : 16]$, for 19 Reynolds numbers, $Re \in [20e^3 : 200e^3]$. Outlier and poststall data was then flagged and removed. This data was then arranged in three-dimensional arrays, flattened, and used to fit a GP-compliant 6-term softmax affine function using the open-source software package *GPfit* [24]. The resulting expression for C_d is log-convex in C_L, Re, and τ, and is remarkably accurate for prestall angles of attack, incurring a mean absolute percentage error of only 1.65%. Slices of this fitted function, the underlying *XFOIL* simulation data, and the resulting model error are displayed in Fig. 8. Note that error becomes large for poststall angles of attack, where the relationship between lift and drag becomes nonconvex in log space. In terms of sensitivity, the pressure drag model is one of the most important constraints in the proposed GP. This 3-D function facilitates the determination of an optimal operating condition (angle of attack can be backed out, if so desired) that is tuned specifically to an optimal wingsail architecture.

Having characterized the aerodynamic models, and static roll moment models, the final component of the roll equilibrium constraint is the hull stability model.

9) Hull Stability: The candidate hull used for wingsail optimization is a 1.78 m fiberglass Santa Barbara Class monohull, outfitted with an 11 kg lead bulbfin keel. A lines drawing of this hull (without the keel and rudder) is provided in Fig. 9, and a corresponding photograph is provided in Appendix B. The righting moment of this hull was first characterized numerically using *Orca3D*, a plugin for *Rhino3D* that provides a suite of naval architectural assessment tools [32]. This approach requires a CAD model of the hull, including the position and mass of the keel, and is thus susceptible to modeling errors. Conveniently, the small size of the candidate hull enabled a direct empirical assessment of the stability curve. This was accomplished by heeling the vessel to various roll angles using a lightweight carbon fiber rod of known length, fixed rigidly to the hull, and measuring the force at the tip of the rod using a conventional load cell. As shown in Fig. 10, the numerical model over-predicted the hull righting moment by a maximum of 16.8% at $\phi = 90°$. This small error likely stemmed from the use of an imperfect CAD model. Nevertheless, numerical estimation of a hull's stability curve is a reasonable option when direct empirical evaluation is not practical.

For most vessels, the hull stability curve is best represented by a sinusoid function, which is not GP-compatible. Formulating the problem as an SP would enable solution sweeps over the desired maximum heel angle; however, in the interests of preserving global optimality, a simplified roll stability constraint was used which requires only the scalar righting moment at a single angle, namely $\phi = 90°$. For a hull with a sealed deck, this ensures universal stability so long as the wingsail is positively buoyant. However, this is a configurable parameter in the proposed GP architecture, and can be adapted to suit other vessel types or sailing scenarios.

10) Mast Location: It is exceedingly common for wingsail designers to position the mast coaxially with the wingsail quarter-chord, which is commonly referred to as the *aerodynamic center:* MacKinnon [33] did so in 1956, Tretow [11] did so in 2017, and so did virtually every other wingsail designer in the intervening eight decades [6], [9], [34]. This choice is motivated by the contention that most of the force generated by a wing is transmitted through its aerodynamic center, and that the moment about this point is precisely zero for symmetric airfoils; however, this common supposition is not always accurate. This subtle misunderstanding is understandable, and likely stems from the fact that thin airfoil theory actually proves this conclusion mathematically; however, *it does so under the assumption of potential flow*. In other words, this theoretical result hinges on the assumption that the airfoil is moving through an incompressible, inviscid fluid, which of course, in practice, it is not. Real airfoils exhibit varying degrees of boundary layer separation near the trailing edge, which effectively alters their shape. For symmetric foils, this flow distortion gives rise to nonzero—and often unstable—pitching moments. This effect is illustrated in Figs. 11 and 21. An unstable pitching moment ($C_m > 0$, $dC_m/d\alpha > 0$) means that the wing's inclination to pitch upward becomes stronger with increasing angle of attack. Consequently, pinning a wingsail at the quarter-chord will not yield the most robust stability characteristics.

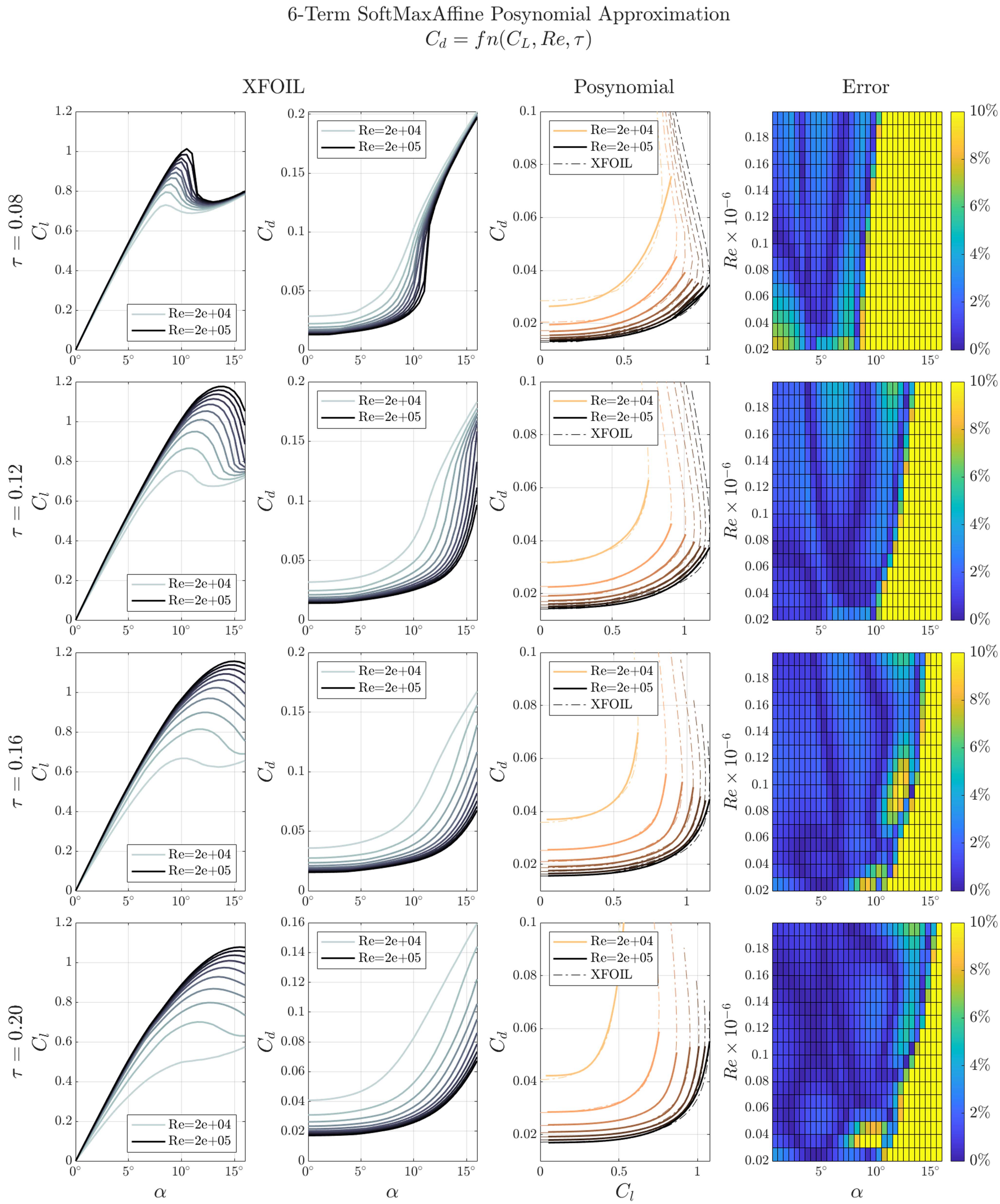

Fig. 8. Posynomial approximation of the profile drag coefficient of symmetric NACA airfoil sections as a function of lift coefficient, Reynolds number, and thickness ratio. Each row corresponds to a single thickness ratio. The two leftmost columns display C_l and C_d at various Reynolds numbers, determined using XFOIL; the third column contains slices of the three-dimensional fitted posynomial function; and the rightmost column displays the posynomial approximation error. The mean absolute percentage error for prestall angles of attack is 1.65%.

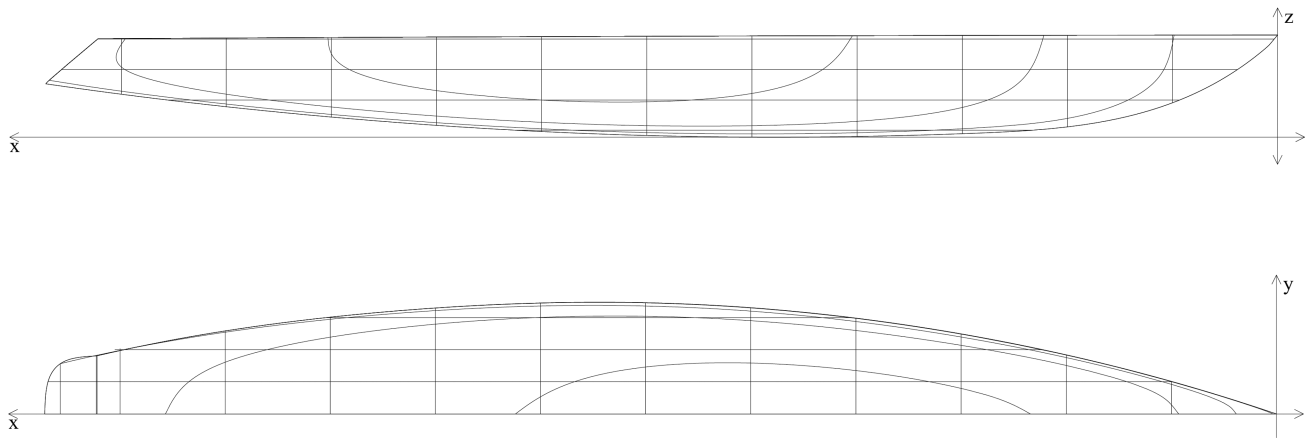

Fig. 9. Body plan and lines drawing for the 1.78 m Santa Barbara Class hull, which features a sealed deck.

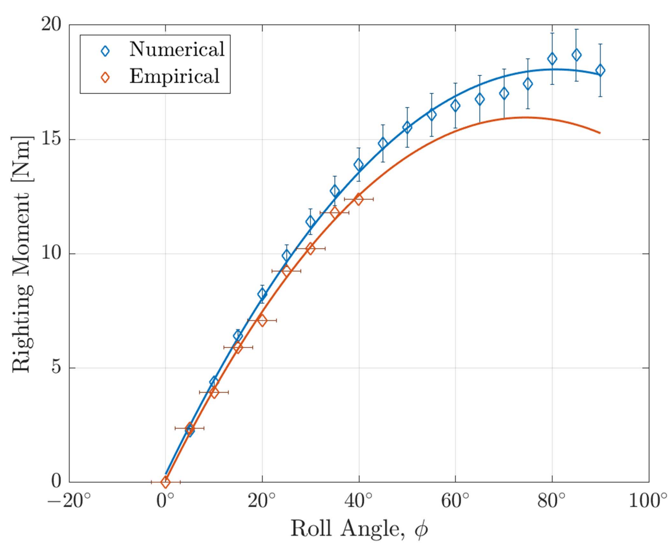

Fig. 10. Stability curve of the candidate hull, determined both numerically and empirically. The vertical error bars on the numerical solution illustrate the impact of a ± 10 mm estimation error of the vertical center of gravity, while the horizontal error bars on the empirical data represent $\pm 3^{\circ}$ of roll angle uncertainty.

The unstable pitching moment induced by boundary layer separation on symmetric foils is quite small, and can be ignored in many instances. Indeed, when using servo-actuated aerodynamic control surfaces, these small moments are treated as disturbances and rejected using feedback control. For servo-actuated aerodynamic control surfaces, open-loop instability may actually be desirable, as it can enable faster setpoint tracking. However, an aerodynamically-actuated wingsail without closed-loop control is extremely sensitive to even minor torque imbalances, and thus must account for these small contributions to the overall moment balance. It should be noted that a small positive moment coefficient on the leading wing will not, in most instances, completely destabilize the full wingsail system; indeed, the restoring torque provided by the tail is generally many times larger, and almost always guarantees system stability [6].

The basic conditions for full system stability are given by the following three requirements:

$$C_{m_{\text{tot}}}(\alpha = 0) = 0 \tag{43}$$

$$C_{m_{\text{tot}}} \leq 0 \tag{44}$$

$$\frac{dC_{m_{\text{tot}}}}{d\alpha} < 0 \tag{45}$$

where $C_{m_{\text{tot}}} = C_{m_W} + C_{m_T} + C_{m_{\text{periph}}}$. As shown in Fig. 11, the wing's contribution to the total moment coefficient is on the order of 0.05; for reference, the tail's contribution is often an order of magnitude larger. Nevertheless, the unstable moment contribution from the wing (and other leading edge peripherals, like anemometers and counterweights) can decrease the overall system damping and exacerbate setpoint overshoot; this can, in turn, provoke a series of stall-recovery hysteresis cycles which decimate aerodynamic performance. This is especially true in gusty, unsteady flows, when the wing must quickly respond to rapid changes in wind speed and direction, in order to avoid stall.

A thought experiment may clarify this phenomenon. Imagine an aerodynamically-actuated wingsail that is initially perfectly aligned with the wind, with no tail deflection. At the next instant, a small deflection is applied to the tail, setting into motion a series of events: first, the tail produces a lift force, which gives rise to a positive moment about the wingsail mast; this aerodynamic torque creates angular velocity, causing the entire wingsail system to rotate clockwise relative to the apparent wind; at this instant, the wing begins generating lift forward of the aerodynamic center, creating a positive (unstable) pitching moment, further increasing the net positive torque on the system; meanwhile, the tail continues to produce positive aerodynamic torque, adding additional angular momentum until its angle of attack is reduced to zero by the rotation of the overall wingsail system; however, the wingsail continues to rotate, driven by its own angular momentum and unstable moment coefficient; at this point, the tail finally begins providing a negative, stabilizing moment; however, it is often too little and too late, and the wing proceeds to stall; the stalled wing produces a strong stabilizing moment, which reduces the pitch angle, and allows

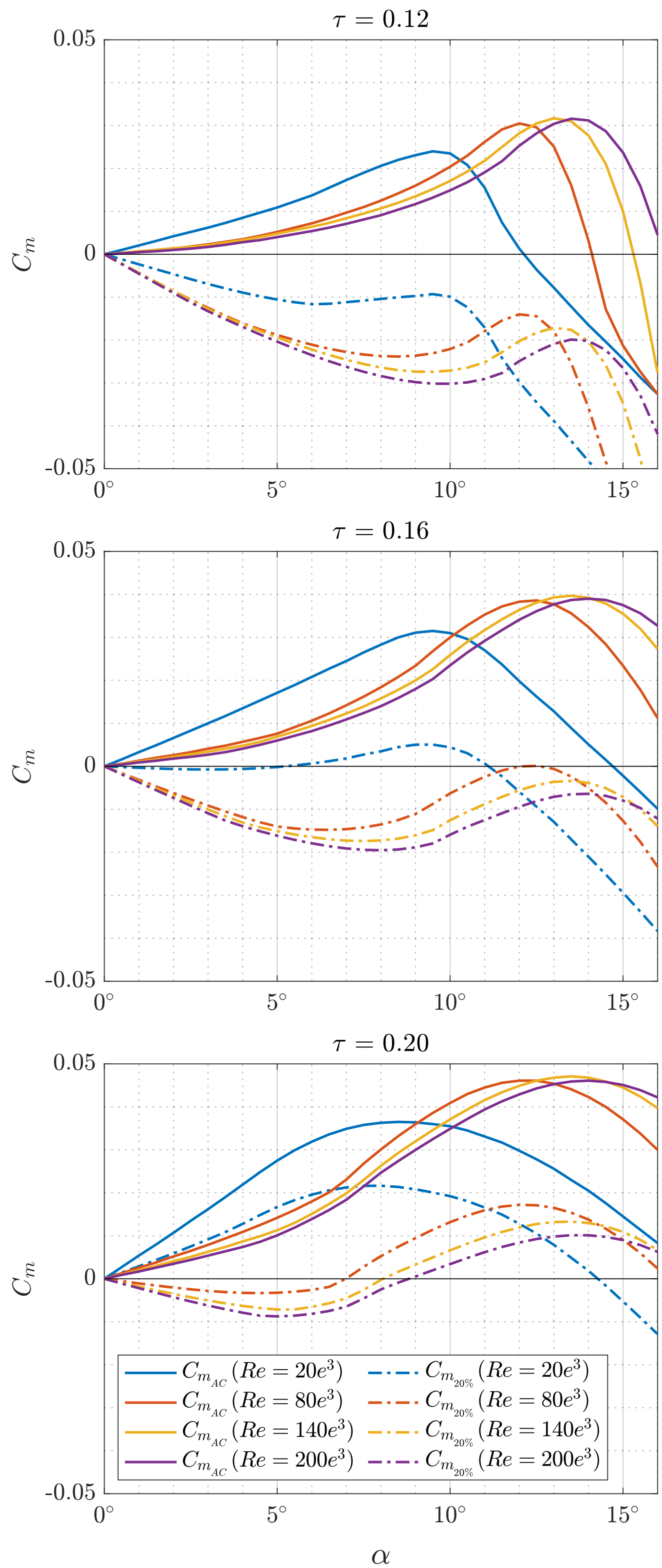

Fig. 11. Moment coefficient polars for three symmetric airfoil sections at various Reynolds numbers, taken with respect to the quarter-chord (solid), and with respect to the 20% chord (dashed). The moment with respect to the 20% chord was computed via integration of the pressure coefficients predicted by *XFOIL*, revealing the chordwise location of the center of pressure, as well as the normal component of the resultant aerodynamic force. Symmetric airfoils exhibit a slightly positive (unstable) moment about the quarter-chord at prestall angles of attack, due to boundary layer separation near the trailing edge; this is particularly true for thicker airfoil profiles.

the wing to recover, but the wingsail is now entrenched in a cyclic stall-recovery pattern.

There is some evidence for the existence of these stall-recovery cycles in the academic literature. In 1962, Blackburn Aircraft constructed an aerodynamically-actuated wingsail pinned at the 30% chord of the leading wing; few details about this platform exist in the public domain, but it has been observed that the vessel exhibited a tendency to develop divergent rolling oscillations and frequently capsized [35]. These oscillations were likely caused by some combination of excessive wingsail weight or moment of inertia, but it is certainly possible that they were exacerbated by stall-recovery cycles. In 1964, Fekete and Newman [36] designed and built an aerodynamically-actuated wingsail which pivoted about the quarter-chord of the leading wing, in order to minimize structural loads. When testing this wingsail, the authors observed that the wing had a tendency to oscillate up to and through stall, noting that the "[wingsail] performance was not as good as had been anticipated...the calculated aerodynamic drag coefficient of the wing was 0.22, indicating a stalled wing on the average."

In light of this finding, it is hypothesized that positioning the mast at the one-fifth chord will provide more robust performance than can be attained by aligning it coaxially with the quarter-chord. This claim is not field-validated in the present work, though it is underpinned by reasonable theoretical analysis. Accordingly, the mast is assumed to be coaxial with the one-fifth chord in the current instantiation of the proposed wingsail GP. Ultimately, however, the position of the mast is a configurable parameter, and does not impact the generality of the proposed wingsail design methodology.

11) Wingsail Balance: In order for the wingsail to be able to maintain a fixed angle of attack when heeled to a nonzero roll angle, it is absolutely vital that it be mass balanced about the mast axis. Even small mass imbalances can drive the wing to stall, reduce the wing angle of attack to zero, or even reverse the angle of attack; this is especially true in light wind, when the aerodynamic restoring forces are small. Of course, any of these outcomes would have disastrous effects on performance. Excluding the roll stability condition outlined earlier, this is perhaps the most important system constraint; indeed, without this constraint, the wingsail will perform so poorly as to be mostly useless for upwind sailing. The mass balance constraint is given by

$$m_{_B} d_{_B} \geq \sum_{i \in \{W,T\}} m_i d_i \tag{46}$$

where $m_{_B}$ and $d_{_B}$ represent the nose ballast mass and distance from the mast axis, respectively, and the right hand side of the equation represents the gravitational moment induced by all wingsail components rear of the mast axis, principally the wing and tail assemblies. To clarify this relationship, and the others that follow in this section, it may be useful to make reference to Figs. 1 and 12, which illustrate the spatial arrangement of the key components in an aerodynamically-actuated wingsail system. This is one instance of a relationship that could be modeled in higher fidelity using an SP formulation; indeed, using the GP-compatible relaxed posynomial equality formulation shown

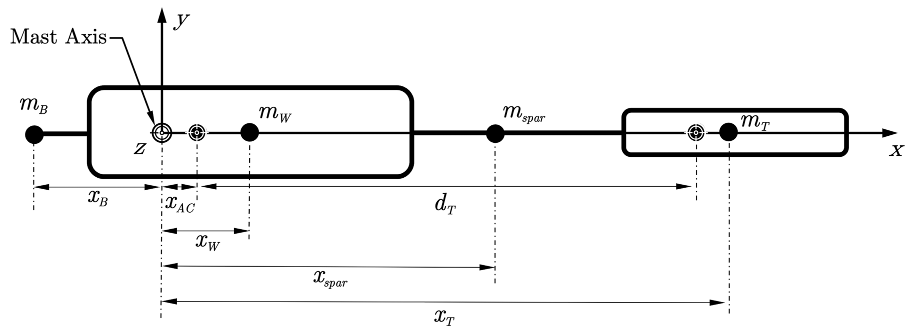

Fig. 12. Wingsail system moment of inertia diagram. In the full GP, the wing and tail are approximated as rectangular prisms, while the nose ballast is represented as a point mass.

in (46), a simplification must be made that the sole contribution to the "nose-down" gravitational moment is the leading edge ballast. As it turns out, this is a reasonable approximation, as m_B typically comprises some 90% of this moment, with only small contributions coming from other leading edge peripherals (i.e., sensors) and the mass of the wing's leading edge itself.

12) Tail Optimization: The tail of an aerodynamically-actuated wingsail system must perform two roles: first, it must provide sufficient aerodynamic "springiness" to permit fast set-point tracking and disturbance rejection; second, it must generate enough aerodynamic torque to hold the wing at a desired angle. These two objectives are at odds: the former favors a tail positioned close to the mast axis, while the latter favors a tail positioned as far from the mast axis as possible. These two objectives must be balanced in a well-reasoned manner.

A simplified wingsail yaw system can be modeled as a second-order linear time-invariant (LTI) system

$$J\ddot{\alpha}_W + B\dot{\alpha}_W = N \tag{47}$$

where α_W is the leading wing angle of attack, J is the total wingsail mass moment of inertia, B is a damping coefficient, N is the total sum of aerodynamic torques applied to the system by the wing, tail, and peripheral appendages

$$\begin{aligned} N &= N_T + N_W + N_P \\ N_T &= -(d_T + x_{AC})F_Y \\ N_W &= \frac{1}{2}\rho U^2 S_W c_W C_{M_W}(\alpha_W, Re, \tau) \\ N_P &= \sum_{i=1}^{N} x_{P_i} \sin(\alpha_W) F_{D_{P_i}}. \end{aligned} \tag{48}$$

Focusing on the tail moment contribution, d_T is the separation between the aerodynamic centers of the wing and tail, and x_{AC} represents the small distance between the mast axis and the aerodynamic center (recall that in this work, $x_{AC} = 0.05\bar{c}_W$). The term F_Y describes the component of aerodynamic tail force that is perpendicular to the leading wing chord line, and can be decomposed in terms of tail lift and drag

$$F_Y = -L_T\cos(\alpha_W) - D_T\cos(\alpha_W). \tag{49}$$

As discussed earlier, and illustrated in Fig. 11, a symmetric wing's contribution to the moment balance (N_W) tends to be slightly positive at prestall angles of attack. Likewise, the drag incurred by physical peripherals placed forward of the mast axis of rotation results in an aerodynamic torque N_P that grows with increasing angle of attack, eroding the stability of the overall system. This influence is rarely discussed in wingsail design, but can be significant for wingsails with long forward spars.

For the purposes of the preceding analysis, a number of simplifications will be made. First, it will be assumed that the influence of the tail is substantially larger than that of the wing and peripheral appendages combined, and only the former component will be treated mathematically. Further, a small angle approximation is applied to (49), so that $F_Y \approx -L_T$. It is also assumed that $d_T + x_{AC} \approx d_T$. Finally, the induced downwash angle felt by the tail is assumed to be small, and is not considered. The reason for these simplifications will soon become apparent, as they enable an elegant and clarifying description of the system dynamics.[8] However, the prudent wingsail designer will do well to keep these oft-forgotten destabilizing influences in mind when it comes time to make final design revisions.

Making reference to Fig. 1, note that $\alpha_T = \alpha_W + \delta_T$, where δ_T is the tail deflection angle, measured clockwise from the leading wing chord line. Substituting the simplified form of (49) into (47), and expanding the L_T term, the following expression is obtained:

$$J\ddot{\alpha}_W + B\dot{\alpha}_W = -d_T\left(\frac{1}{2}\rho U^2 S_T\right)\left[C_{L_{\alpha_T}}(\alpha_W + \delta_T)\right] \tag{50}$$

where $C_{L_{\alpha_T}}$ is the linear prestall lift slope constant. By setting $k = 0.5 d_T \rho U^2 S_T C_{L_{\alpha_T}}$, and rearranging the equation algebraically, a second-order ordinary differential equation in standard form is obtained

$$\ddot{\alpha}_W + \left(\frac{B}{J}\right)\dot{\alpha}_W + \left(\frac{k}{J}\right)\alpha_W = -\left(\frac{k}{J}\right)\delta_T. \tag{51}$$

Note that the constant k represents a sort of aerodynamic spring constant; however, it is important to note that this formulation hinges on an assumption of linearity: namely, that the scalar

[8] In optimization, one must balance generality and efficiency; in mathematical modeling, one must balance fidelity and interpretability.

lift slope parameter remains constant for all angles of attack, and that lift increases without bound with increasing angle of attack. Of course, this is absolutely not the case in practice, due to the limitations imposed by boundary layer separation and stall. Nevertheless, for small excursions from the zero-lift angle of attack, it is a useful approximation, and elucidates the system response to tail deflection commands. With the system dynamics expressed in this manner, it is clear that the system's undamped natural frequency is given by

$$\begin{aligned}\omega_n &= \sqrt{\frac{k}{J}}\\ &= \sqrt{\frac{\frac{1}{2} d_T \rho U^2 S_T C_{L_{\alpha_T}}}{J_B + J_W + J_T}}. \end{aligned} \tag{52}$$

This expression warrants inspection, as it reveals the key parameters which influence the open-loop bandwidth of the wingsail system. Indeed, it is now apparent that ω_n increases linearly with wind speed,[9] and with the square-root of tail area (S_T) and tail separation distance (d_T). It is desirable for the natural frequency to be as large as possible, in order to permit fast tracking of fluctuations in wind direction, and in doing so avoid stalling the wing, tail, or both. Additional insight can be gleaned by simplifying the denominator.

First, it is worth acknowledging that the nose ballast and tail moments of inertia, J_B and J_T, typically account for at least 15% and 70% of the overall system moment of inertia, respectively. The dominance of these terms is explained by the distance-squared term in the Parallel Axis Theorem, which states that moment of inertia of an object with respect to any given axis is equal to the moment of inertia with respect to its own centroidal axis, plus the product of its mass and the square of the distance between the two axes. Thus, it is reasonable to approximate the nose and tail as point masses (ignoring the moment of inertia about their own respective centroidal axes), and ignore the wing moment of inertia entirely, giving a simplified formulation of wingsail natural frequency

$$\omega_n = \sqrt{\frac{\frac{1}{2} d_T \rho U^2 S_T C_{L_{\alpha_T}}}{m_B x_B^2 + m_T x_T^2}}. \tag{53}$$

Recall that the wing must be balanced, such that the constraint presented in Section III-C11 is satisfied. This gives the following approximation: $m_B \approx m_T d_T / d_B$. Substituting this expression in (53) yields an interpretable formulation of wingsail natural frequency

$$\omega_n = \sqrt{\frac{\frac{1}{2} \rho U^2 S_T C_{L_{\alpha_T}}}{m_T (x_T + x_B)}}. \tag{54}$$

This simplification provides yet more insight: in order to maximize the natural frequency, one must maximize the ratio of tail area to tail mass, while minimizing the distance of both the tail and nose ballast from the mast axis. As usual, these objectives are inhibited by other objectives and constraints. For one, reduction of nose ballast *distance* requires a commensurate increase in nose ballast *mass* in order to preserve the constraint presented in Section III-C11. There is no upper bound to this relationship: blind optimization for high natural frequency (subject to a balance constraint) will drive the nose ballast separation to zero, and the nose ballast mass to infinity, thus overwhelming the finite reserve buoyancy of the hull. Likewise, single-minded prioritization of natural frequency will reduce the tail separation as much as possible, placing the tail in the wake of the wing.

Ultimately, however, the proposed GP *does not include natural frequency in the objective function.* This is perhaps the greatest weakness of the proposed approach: maximization of *steady-state* lift-over-drag does nothing to ensure that the wing will not routinely stall in a dynamic wind field. And while the objective function could be formulated as a weighted sum of lift-over-drag and natural frequency, there is no natural, physics-informed means of selecting the corresponding weights. Accordingly, constraints are developed to reflect the understanding that a truly optimal wingsail must posses both of these qualities.

First, the tail should be placed as close as possible to the mast axis, without placing it in the wake of the wing; for while the aerodynamic spring factor increases with tail separation, the moment of inertia term increases with tail separation squared. Wake dynamics can be difficult to characterize, but a reasonable constraint would ensure that the tail maintains a separation distance of at least one wing-chord length from the trailing edge of the wing root [37]. Expressed in terms of the wing and tail parameters, this constraint takes the form

$$d_T \geq \frac{3}{4} c_{r_W} + \bar{c}_W + \frac{1}{4} c_{r_T}. \tag{55}$$

Ideally, it would be possible to formulate a relaxed posynomial equality constraint that ensures the total wingsail mass remains below the reserve buoyancy of the hull; however, this relationship does not meet the requirements for relaxation set out in Section II. Accordingly, the nose ballast position constraint attempts to capture the Pareto-optimal tradeoff between low mass and low moment of inertia. For the candidate hull, this position approximately coincided with the half-beam of the vessel, which also reduces the likelihood of wave impact on the nose ballast. In terms of vertical position, the nose ballast should be placed as close to the deck as possible, as there is no advantage to elevating it. These observations yield two additional constraints on nose ballast position

$$d_B \geq \frac{b_{\text{hull}}}{2} \tag{56}$$

$$z_B = z_{r_W}. \tag{57}$$

Having discussed the tail system objectives from the dynamics perspective, it is now time to analyze the static holding torque provided by the tail at steady state. The static moment balance between the wing and the tail is given by

$$\begin{aligned} d_T F_{Y_T} &\geq N_W(\alpha_{W_{\max}}) \\ d_T[-L_T \cos(\alpha_{W_{\max}}) - D_T \sin(\alpha_{W_{\max}})] &\geq N_W(\alpha_{W_{\max}}) \\ \frac{d_T}{\bar{c}_W}\frac{S_T}{S_W}[-C_{L_T}\cos(\alpha_{W_{\max}}) - C_{D_T}\sin(\alpha_{W_{\max}})] &\geq C_{N_W}(\alpha_{W_{\max}}). \end{aligned} \tag{58}$$

[9]This explains, in part, why aerodynamically-actuated wingsails tend to exhibit better performance at higher wind speeds.

For upwind sailing, holding torque is typically not an issue, as not much is required to hold the wing at small angles of attack; however, for downwind sailing—when it is advantageous to hold the wing at large angles of attack in excess of 40°, so called *deep stall*—this is not the case. A wing in deep stall effectively behaves like a flat plate, with the center of pressure moving rearward toward the trailing edge, driving the wing to reduce its pitch angle. The moment coefficient of a flat plate in deep stall is given by

$$C_{N_{\text{plate}}} = -0.4 \sin(\alpha_{\text{plate}}) \qquad \alpha_{\text{plate}} \in [30^\circ : 150^\circ] \tag{59}$$

where α_{plate} is the angle of attack of the plate. Recognizing that the term $(d_T/\bar{c}_W)(S_T/S_W)$ is the aerodynamic tail volume (V_T), and applying first and second order Taylor series expansions of sine and cosine, respectively, and setting $\alpha_{W_{\max}} = 45^\circ$, (58) can be expressed as a posynomial inequality constraint

$$C_{L_T} \geq \frac{1}{2} C_{L_T} \left(\frac{\pi}{4}\right)^2 + C_{D_T}\left(\frac{\pi}{4}\right) + \frac{0.2828}{V_T}. \tag{60}$$

An advantage of utilizing total lift-over-drag as the GP cost function is that it automatically promotes low-drag tail designs. While it is advantageous for the forward wing to operate at moderate to high lift coefficients, the tail should operate at the lowest possible lift coefficient which stabilizes the forward wing about a desired angle of attack. As a consequence, the proposed GP tends to promote thinner, larger tails which operate at lower C_L (i.e., lower angle of attack). This design incurs less induced drag, and is more conservative from a dynamics perspective: in unsteady flow conditions, it is advantageous to maintain a small angle of attack on the tail to reduce the likelihood of stall.

IV. Optimal Taper Ratio for Boundary Layer Flow

The following section describes an investigation into the optimal taper ratio for a wing subject to a logarithmic spanwise velocity profile. This specific vertical velocity profile is well-known to describe the atmospheric boundary layer, up to roughly 2000 m above the Earth's surface. This research effort enabled the creation of a GP-compatible posynomial surrogate function that expresses the optimal taper ratio as a function of wingspan, wing root elevation above the free surface, and a single parameter which adjusts the shape of the atmospheric boundary layer.

A. Elliptical Lift and Lift-Induced Drag

Lift-induced drag is a 3-D effect caused by fluid flow around the tips of finite-span wings. This flow produces characteristic wingtip vortices which reduce both the effective wingspan and local attack angle of the wing, leading to an increase in drag, and a slight reduction in lift. Thus, induced drag reduces the efficiency (i.e., lift-over-drag ratio) of finite-span wings. One of the most celebrated findings of lifting line theory is that this deleterious effect is minimized for wings with an elliptical spanwise lift distribution [38].

A full derivation of this result is beyond the scope of this manuscript, but a brief description is provided here for completeness: first, a virtual wing is described using an array of discrete vortices in a mean flow, in accordance with the Kutta–Joukowski theorem; next, the 2-D circulation at each spanwise station is expressed as a Fourier series with unknown coefficients; this function is then integrated over the full wingspan to estimate the local induced vertical velocity at each station, which, together with free-stream velocity, determines the local induced downwash angle; having related the local circulation to downwash angle, it is then possible to solve a finite system of equations to determine the Fourier coefficients, yielding expressions for lift and lift-induced drag

$$L = 2\pi b^2 \rho U^2 A_1 \tag{61}$$

$$D_i = 2\pi b^2 \rho U^2 \sum_{n=1}^{\infty} n A_n^2 \tag{62}$$

where $A_1 = SC_L/(4\pi b^2)$ is the first coefficient of the Fourier series approximation, and determines the lift produced by the airfoil (the other coefficients alter the distribution of lift along the span, but do not affect the total lift). The induced drag is thus minimized when all other Fourier coefficients besides A_1 in the series are zero; in this case, the distribution of circulation across the wingspan is elliptical, and given by

$$\Gamma(\eta) = 4bUA_1\sqrt{1-\eta^2} \tag{63}$$

where $\eta \in [0:1]$ is the nondimensional spanwise coordinate.

Since this finding was first published over a century ago, aircraft and yacht designers alike have gone to great lengths to create efficient wings which approximate this advantageous lift profile. There are numerous ways to accomplish this goal, including the use of different airfoil profiles along the span, wing twist (or *washout*), and adopting an elliptical planform.

The use of an elliptical planform is arguably the simplest means of attaining an (approximately) elliptical lift profile; however, in practice, such a planform is difficult to manufacture: both the leading and trailing edges are curved, which makes skinning the wing challenging. Moreover, as one moves from the wing root to the tip, the aerodynamic center shifts toward the trailing edge, which complicates structural design. Fortunately, the latter issue can be ameliorated using a pair of quarter-ellipses, sized such that the semi-major axis of each is colinear with the quarter-chord line. The semi-minor axes are thus constrained by the desired root chord, according to the expression $c_r = c_1/4 + 3c_2/4$, where c_1 and c_2 are the leading edge and trailing edge quarter-ellipses, respectively. This is a time-tested solution, and was famously implemented on the British Supermarine Spitfire. However, the first problem remains: it is difficult to manufacture a wing with continuous planform curvature. It is thus desirable to determine an optimal linear approximation of the elliptical planform.

B. Optimal Taper Ratio for Uniform Flow

A tapered wing is characterized by a linear reduction of the wing chord from root to tip, and provides a compromise between the aerodynamic efficiency of an elliptical wing and manufacturability of a rectangular wing. Such a wing is also easier to describe mathematically. As a preliminary exercise, we will use convex optimization to determine the optimal taper ratio $\lambda = c_r/c_t$ assuming uniform flow across the wing (in other words, assuming an elliptical planform will yield an elliptical lift profile). First, we describe the ideal chord distribution as a function of spanwise position, in nondimensional form

$$\tilde{c}^*(\eta) = \sqrt{1-\eta^2}. \tag{64}$$

Next, we use the equation of a generic line to define a linear, straight-tapered planform in terms of the root chord c_r, taper ratio λ, and the nondimensional spanwise coordinate η

$$c(\eta, \lambda, c_r) = c_r[1-(1-\lambda)\eta]. \tag{65}$$

We then compute the squared difference between (65) and (64), and integrate the result along the spanwise coordinate η

$$\mathcal{C}_\lambda = \int_0^1 \frac{1}{2}(c-\tilde{c}^*)^2 d\eta. \tag{66}$$

Differentiating the cost function with respect to the two free variables, c_r and λ, and setting the resulting equations equal to zero yields a system of two equations with two unknowns. We find that the taper ratio that best approximates an elliptical planform is given by

$$\boxed{\lambda^* = \frac{4-\pi}{2(\pi-2)} = 0.376}. \tag{67}$$

This result has been extensively corroborated by both theory and empirical data sets. In 1947, Glauert [30] noted, "It appears that the best results are obtained when the tip chord is from one-third to one-half the central chord..."; later, in 1965, Hoerner [39] observed, "...for taper ratios between 0.3 and 0.4, the additional drag is very small, on the order of 1 or 2% in aspect ratios commonly employed on airplanes". However, it is important to note that both of these observations are based upon a fundamental assumption: spanwise uniformity of the free-stream flow.

C. Sail Taper

Aircraft wings and wingsails are differentiated principally by the fact that the latter are subject to a significant spanwise wind velocity gradient, imposed by the Earth's atmospheric boundary layer. This effect is particularly severe for smaller vessels outfitted with high aspect ratio wingsails, due to the fact that the average horizontal wind velocity in the boundary layer grows logarithmically with elevation. Surprisingly, given the immense body of research motivated by sailboat racing [26], [40], [41], [42], and sail-assisted oceanic cargo transport [43], [44], [45], relatively little attention has been paid to the effect of the Earth's boundary layer on sail forces.

The first formal mathematical treatment of a vertical wing subject to a spanwise velocity gradient was made by Milgram [46] in 1973. In this work, lifting-line theory was extended by assuming a *linear* spanwise gradient of the form $U(z) = U_0 + kz$. This approach is mathematically elegant, and yielded more expressive formulations of lift and induced drag; however, it is well known that the atmospheric boundary layer is not in fact linear, but rather logarithmic [47]. In the decades which followed this seminal work, very little attention was paid to logarithmic spanwise gradients. This conspicuous deficiency in the academic literature was noted by Xiangming and Huawu [48], who conducted an aerodynamic analysis of an unconventional elliptic arc wingsail using a wind tunnel tuned to mimic the Earth's atmospheric boundary layer. Their experiments indicated that a spanwise velocity gradient can reduce both the lift and drag coefficient of a wingsail by 16% –44% and 11% –42%, respectively. More recently, Li et al. [49] published a series of CFD simulations supporting the opposite conclusion—that a spanwise velocity gradient actually *increases* the maximum lift coefficient, and delays stall by compressing the separated vortex. While illuminating, it is not clear whether this result generalizes to other wing geometries, boundary layer shapes, and Reynolds number regimes. All simulations assumed $Re = 2.4 \times 10^6$, a regime attainable only by the fastest competitive sailing vessels. Further, the authors assume a power-law wind profile, which—while easier to integrate than a logarithmic wind profile—does not account for surface roughness, and is known to be unrepresentative of the true atmospheric boundary layer in most cases [50].

While the quantitative impact of a logarithmic spanwise velocity gradient on a wing remain unclear, the preceding analyses suggest that it likely exerts some influence on performance. This contention is corroborated by the design of traditional soft sails, most of which incorporate a substantial degree of spanwise wing twist, or *washout*, to reduce the section lift coefficient near the masthead, where higher wind speeds are expected. This produces a distribution of lift along the span that more closely resembles the elliptical ideal. Although washout cannot be incorporated into a rigid wingsail without the use of multiple independent trailing edge flaps, the taper ratio can be adjusted to account for the boundary layer flow, and thus better approximate an elliptical lift profile.

D. Optimal Taper Ratio in Logarithmic Flow

To the best of the author's knowledge, the optimal taper ratio for a wing subject to a logarithmic spanwise gradient has not been determined. This is not entirely surprising, given that such a scenario pertains almost exclusively to the domain of sailing. Indeed, while various aerodynamic lifting surfaces are exposed to linear velocity gradients (i.e., helicopter and propeller blades), only the sailing vessel is exposed to meaningful logarithmic spanwise velocity gradients. Another potential explanation for this void in the academic literature is that an analytical solution may not exist: the ratio of logarithmic functions used to characterize the atmospheric boundary layer is extremely unamenable to integration. In the following sections, optimal taper ratio for a wing exposed to a logarithmic spanwise velocity profile is derived via numerical integration.

The logarithmic wind profile, defined in (68) and displayed in Fig. 13, is a semi-empirical relationship that describes the change in average horizontal wind speed with vertical distance

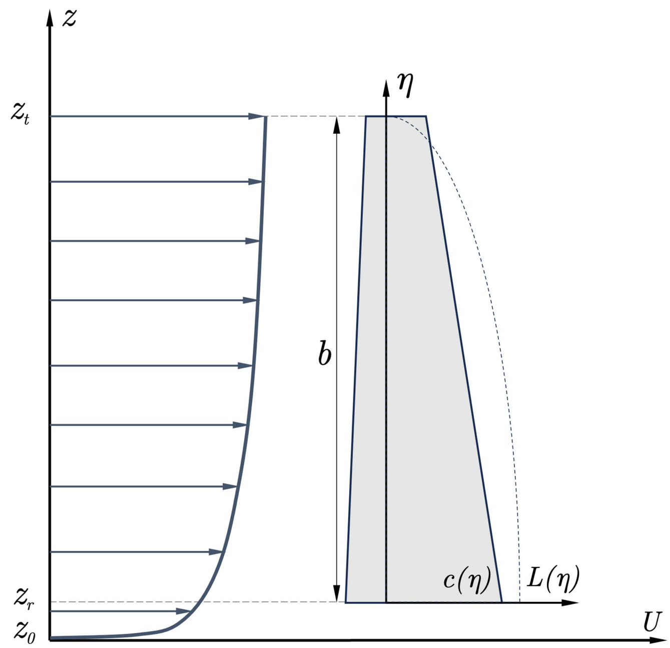

Fig. 13. Coordinate system definitions for log layer analysis. The velocity distribution on the left represents the atmospheric boundary layer, $U(z)$, while the thin dotted line plotted on the inset coordinate frame represents an elliptical lift profile, $L(\eta)$, where η is defined as the normalized spanwise dimension.

above the Earth's surface

$$U(z) = U_r \frac{\ln\left(\frac{z}{z_0}\right)}{\ln\left(\frac{z_r}{z_0}\right)}. \tag{68}$$

This equation permits the prediction of the average wind speed at an elevation of interest, given a known reference wind speed (U_r) at a known reference elevation (z_r), along with the surface roughness (z_0). The surface roughness is expressed in meters, and influences the structure of the velocity gradient.

In oceanographic and meteorological fields, the most common reference elevation is 10 m; however, for the purposes of the following derivation, the chosen reference elevation is the wingsail wing tip height above the free surface, z_t. Substitution of this reference elevation into (68) yields an expression for average wind velocity as a function of nondimensional spanwise position along a vertically-oriented wing

$$U(z) = U_t \frac{\ln\left(\frac{z}{z_0}\right)}{\ln\left(\frac{z_t}{z_0}\right)} \tag{69}$$

$$U(\eta) = \frac{\ln\left(\frac{b\eta+z_r}{z_0}\right)}{\ln\left(\frac{b+z_r}{z_0}\right)}. \tag{70}$$

Recall that the sectional lift generated by an airfoil section is given by

$$\begin{aligned} l(\eta) &= \frac{1}{2}\rho U(\eta)\Gamma(\eta) \\ &= \frac{1}{2}\rho U(\eta)^2 c(\eta) C_l. \end{aligned} \tag{71}$$

In order to minimize induced drag, the shape of this lift function over the wingspan must be elliptic, taking the form

$$l^*(\eta) = l(0)\sqrt{1-\eta^2} \tag{72}$$

where $l(0) = 0.5\rho U_r^2 c_r C_l$. Substituting (70) into (71), and equating (71) and (72) gives an expression, which can be solved to determine the optimal chord distribution for a wing exposed to a logarithmic spanwise velocity profile

$$\frac{1}{2}\rho\left[U_t \frac{\ln\left(\frac{b\eta+z_r}{z_0}\right)}{\ln\left(\frac{b+z_r}{z_0}\right)}\right]^2 c(\eta)C_l = \frac{1}{2}\rho U_r^2 c_r C_l \sqrt{1-\eta^2} \tag{73}$$

$$\begin{aligned} c^*(\eta) &= \left(\frac{U_r}{U_t}\right)^2 \left[\frac{\ln\left(\frac{b+z_r}{z_0}\right)}{\ln\left(\frac{b\eta+z_r}{z_0}\right)}\right]^2 c_r\sqrt{1-\eta^2} \\ &= \left[\frac{\ln\left(\frac{z_r}{z_0}\right)}{\ln\left(\frac{b+z_r}{z_0}\right)}\right]^2 \left[\frac{\ln\left(\frac{b+z_r}{z_0}\right)}{\ln\left(\frac{b\eta+z_r}{z_0}\right)}\right]^2 c_r\sqrt{1-\eta^2} \\ &= \left[\frac{\ln\left(\frac{z_r}{z_0}\right)}{\ln\left(\frac{b\eta+z_r}{z_0}\right)}\right]^2 c_r\sqrt{1-\eta^2}. \end{aligned} \tag{74}$$

Equation (74) describes the optimal planform shape for a wing with span, b, root chord, c_r, and root elevation, z_r, operating in a boundary layer flow with roughness length z_0. This relationship can be simplified by introducing the following nondimensional quantities:

$$\tilde{c}(\eta) = \frac{c(\eta)}{c_r} \tag{75}$$

$$\lambda = \frac{c_t}{c_r} \tag{76}$$

$$k_1 = \frac{b}{z_r} \tag{77}$$

$$k_2 = \frac{z_r}{z_0} \tag{78}$$

$$\boxed{\tilde{c}^*(\eta) = \left[\frac{\ln(k_2)}{\ln\left[k_2(k_1\eta+1)\right]}\right]^2 \sqrt{1-\eta^2}}. \tag{79}$$

It is perhaps illustrative to pause for a moment and inspect this equation. The smaller the relative vertical separation between wing root and tip (in other words, as k_1 becomes small), the more closely the boundary-layer-optimal planform resembles the standard elliptical planform. This is because the logarithmic velocity ratio term approaches unity as k_1 approaches zero, implying that both the root and the tip are exposed to the same velocity. However, when the difference between the root and tip becomes large, the wing extremities are exposed to drastically different velocities, and the logarithmic velocity ratio term dominates the square-root elliptical planform term for small η (i.e., near the root); consequently, the ideal ellipsoidal lift planform exhibits a progressively more dramatic concavity change as k_1 becomes large. Interestingly, absolute velocity itself does not enter into the equation at all; this is because exponential and

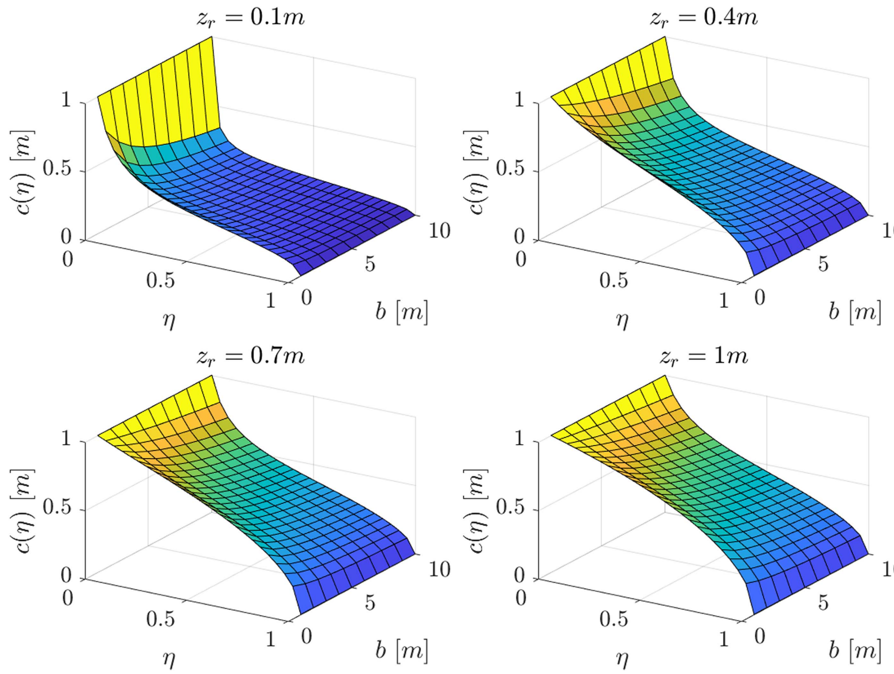

Fig. 14. Illustration of the influence of wing root elevations (z_r) and wingspan (b) on the optimal planform shape ($c(\eta)$) for a wing exposed to a logarithmic spanwise velocity profile. The range of parameter values was chosen in light of the typical rig dimensions of small sailing vessels.

logarithmic functions are self-similar, and the absolute values do not matter—only the relative values (i.e., how the tip velocity relates to the root velocity) matter. This result is quite dramatic: when the wing root is very close to the free surface, and the wing tip is many times higher, the ideal planform is "cobra-shaped," with a massive root chord that rapidly tapers off, and then stays mostly constant, as displayed in Fig. 14.

Nondimensionalization of this expression is absolutely vital; not only does it clarify the relationships involved, but it also reduces the dimensionality of the problem. Without this dimensional reduction, direct numerical integration of the error function used to find the optimal taper ratio would be infeasible.

Recall that the nondimensional expression for the chord distribution of a single-taper wing is given by (65). This equation can be subtracted from (79) to obtain an error function; this function can then be integrated numerically to obtain a cost function. It is then possible to determine the taper ratio which minimizes this function, for a given set of parameters k_1 and k_2

$$E(k_1, k_2, \lambda, c_r) = \Bigg| c_r[1 - (1 - \lambda)\eta] - \left[\frac{\ln(k_2)}{\ln[k_2(k_1\eta + 1)]}\right]^2 \sqrt{1 - \eta^2}\Bigg| \tag{80}$$

$$[\lambda^\star, c_r^\star] = \arg\min_{\lambda, c_r}(E) = fn(k_1, k_2). \tag{81}$$

For all combinations of k_1 and k_2 considered, the optimal single taper was able to replicate the ideal elliptical lift distribution with a root mean squared percentage error of 4.45%, and a maximum error of 7.4%. A representative sample of these optimal planforms are presented in Fig. 15. However, for this result to be of use in a GP optimization context, the optimal taper ratio function must be described using a surrogate posynomial function. This function was estimated in a similar manner as the pressure drag relationship described in Section III, using a 3-term softmax affine function. The resulting fitted function is displayed in Fig. 16, and exhibits a relatively large percentage root mean squared error of 10.2%, over the parametric domain considered; however, this error is dominated by corners of the function corresponding to unlikely combinations of the nondimensional parameters k_1 and k_2—for example, describing sail rig configurations with very short wingspan, but very high wing root.

E. Roughness Length

Selection of an appropriate boundary layer roughness length should be conducted on the basis of the typical wind and sea states in which the wingsail will be deployed. A number of roughness length models have been proposed; for the present work, the COARE model described by by Edson et al. [47] was chosen for its accuracy and ability to relate roughness length directly to the U_{10} reference wind speed. In this parameterization, roughness length is decomposed into smooth and rough components, both of which are a function of the friction velocity (u_*)

$$\begin{aligned} z_0 &= z_0^{\text{smooth}} + z_0^{\text{rough}} \\ z_0^{\text{smooth}} &= \gamma \frac{\nu}{u_*} \\ z_0^{\text{rough}} &= \alpha \frac{u_*^2}{g} \end{aligned} \tag{82}$$

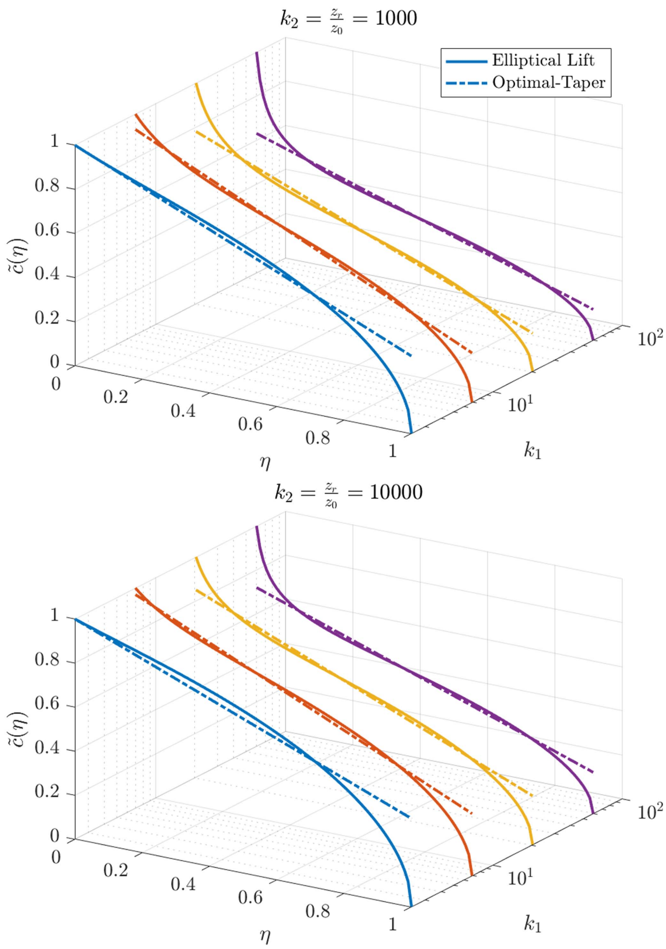

Fig. 15. Optimal nondimensional single-taper planforms for a wing exposed to a logarithmic spanwise velocity profile.

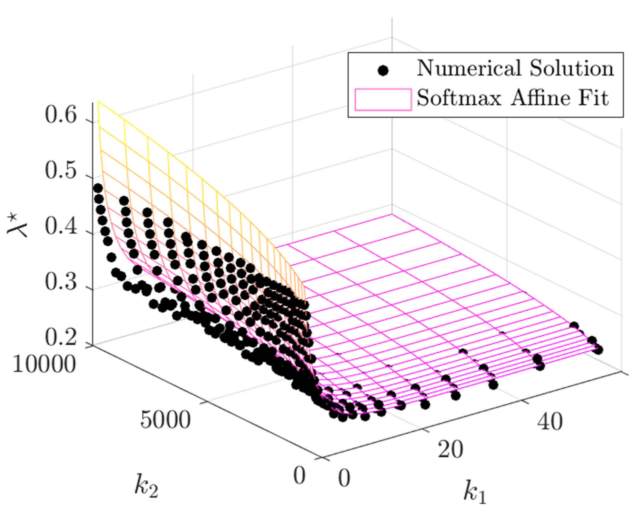

Fig. 16. GP-compatible softmax affine function describing the optimal taper ratio for a wing subject to a logarithmic spanwise velocity profile. The root mean squared percentage error over the entire domain is 10.2%, though this is largely driven by corner-case wingsail configurations that are unlikely to occur in practice.

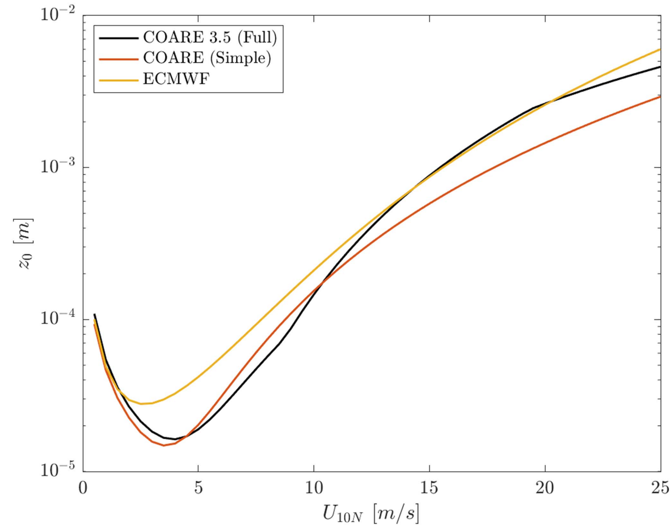

Fig. 17. Boundary layer roughness length as a function of 10 m reference wind speed, determined using three different models. In the proposed GP formulation, the full COARE 3.5 model is used to estimate the roughness length.

where γ is the roughness Reynolds number for smooth flow, which has been determined to be 0.11 from laboratory experiments; α is the Charnock coefficient, which represents normalized roughness; u_* is the friction velocity; $\nu = 14.88e^{-6}\mathrm{m}^2/\mathrm{s}$ is the kinematic viscosity of air; and $g = 9.81\mathrm{m}/\mathrm{s}^2$ is gravitational acceleration. According to the COARE 3.5 model, the Charnock coefficient can be estimated using the following expression:

$$\begin{aligned} \alpha &= \min[\alpha^{\text{slow}}, \alpha^{\text{fast}}] \\ \alpha^{\text{slow}} &= 0.0017U_{10} - 0.005 \\ \alpha^{\text{fast}} &= 0.028. \end{aligned} \tag{83}$$

The friction velocity, likewise, can be estimated using the following expression:

$$\begin{aligned} u_* &= \max[u_*^{\text{slow}}, u_*^{\text{fast}}] \\ u_*^{\text{slow}} &= 0.03U_{10} \\ u_*^{\text{fast}} &= 0.062U_{10} - 0.28. \end{aligned} \tag{84}$$

Substituting (83) and (84) into (82) results in a formulation of the roughness length as a direct function of the 10 m reference wind speed, as displayed in Fig. 17. For a reference wind speed $U_{10} = 10$ m/s, the roughness length is $1.545e^{-4}$ m, and this is the value that is used in the GP for the purposes of wingsail design.

V. Results

The GP approach to wingsail design optimization is an incredibly flexible framework; it provides not only the optimal solution to the GP as formulated, but also a vast number of *optimal-adjacent* design permutations, using *design sweeps*. Design sweeps constrain one or more decision variables to a set of predefined discrete scalar values, and repeatedly solve

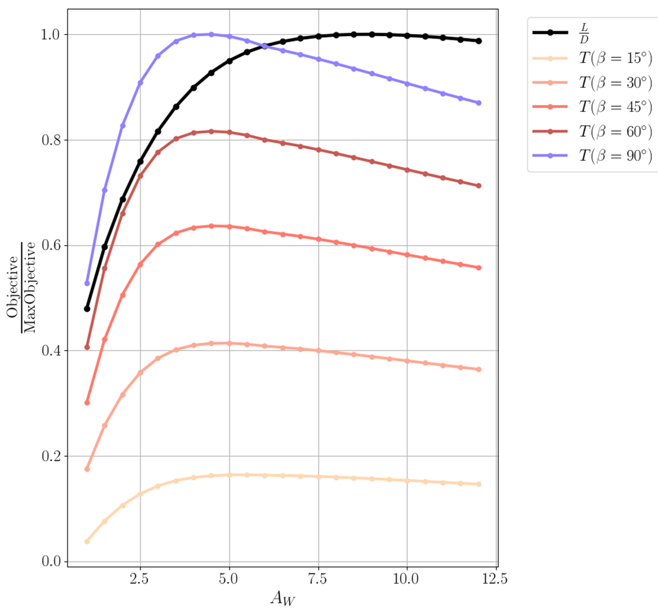

Fig. 18. Thrust is not a GP-compliant posynomial; however, it can be computed offline after the GP is solved, permitting comparison with the GP objective function. The optimal wing aspect ratio with respect to lift-over-drag is $A_W = 9$; however, thrust is maximized in the vicinity of $A_W = 4.5$, for all but the highest pointing angles (β). This design sweep underscores the value in computing optimal-adjacent solutions.

the GP, in order to assess how the objective function changes for off-optimal values of the decision variable in question. Each discrete point in a variable sweep represents a distinct full-system solution. Depending on the sensitivity of the objective function to the particular parameter of interest, these optimal-adjacent solutions may yield a better balance between opposing influences than the numerically optimal solution. In other words, they may be optimal in a *multiobjective* sense. Moreover, design sweeps permit the inspection of non-GP-compatible relationships computed offline after optimization.

For example, the proposed wingsail GP strives to maximize *lift-over-drag* as a proxy for *upwind thrust*. Thus, the GP provides solutions that are globally optimal with respect to lift-over-drag, but not necessarily with respect to thrust. The error incurred by this approximation is illuminated by investigating optimal-adjacent solutions, as shown in Fig. 18. By holding aspect ratio fixed at various values, and iteratively resolving the GP, it becomes clear that the optimal aspect ratio with respect to lift-over-drag ($A_W = 9$) does not yield the maximum thrust, for all but the most severe (small) upwind pointing angles. Indeed, for moderate upwind pointing angles ($\beta = 45°$), the use of a much lower aspect ratio wing ($A_W = 4.5$) results in an 6.8% increase in thrust.

It should be noted that neither thrust nor lift-over-drag are particularly sensitive to aspect ratio in the range $A_W \in [4 : 10]$. Once an acceptable parameter range has been delineated, the final system configuration can be determined based on other considerations, such as natural frequency. For instance, although reducing the leading wing aspect ratio from $A_W = 9$ to $A_W = 4.5$ provides a slight increase in thrust, it also incurs a tremendous 117% increase in the system moment of inertia, a 35.4% increase in overall mass, and a 16.5% reduction in natural frequency. Therefore, although the lower aspect ratio design is capable of providing more thrust at steady state, it may stall more frequently in unsteady wind conditions.

These tradeoffs underscore the true utility of proposed design methodology: it provides design *insights*, rather than simply returning a single ostensibly optimal *solution*. The design of sailing rigs is invariably a multiobjective optimization problem. One must balance survivability and speed; cost and complexity. There is no such thing as a single optimal wingsail. The final system configuration will reflect the specific needs of the designer, which may include: longevity, endurance, performance, or some combination thereof. The proposed wingsail design optimization methodology is designed to provide guidance and clarity, not (necessarily) final answers.

Table I displays a summary of the optimal-adjacent GP solution obtained for $A_W = 4.5$. The full solution contains 121 free variables in 203 constraints, and was solved in 0.079 s on a laptop computer. The chord Reynolds number and aerodynamic coefficients are solved separately for the wing ($Re_W = 87,650$) and the tail ($Re_T = 63,710$), assuming a reference velocity, $U_{10} = 8$ m/s, and an atmospheric boundary layer roughness length, $z_0 = 1.545e^{-4}$ m. This boundary layer shape results in a wind velocity $U = 5.86$ m/s at the elevation of the leading wing's aerodynamic center of effort, $z_{\text{CE}} = 0.724$ m.

VI. Discussion

The solution to a GP is guaranteed to be globally optimal with respect to the objective function and constraints; however, it is argued that while the single globally optimal solution can provide useful and interesting design guidance, of greater value is the *sensitivity* information returned with each GP solution. Indeed, GP sensitivity illuminates the shape of the objective function near the global minimum, which elucidates the general nature of the underlying parameter space. There are multiple ways to assess sensitivity, one being the use of design sweeps, as described in Section V. Variable sweeps can also be used to reveal how other, nonobjective decision variables vary relative to one another. For example, Fig. 19 shows how total lift increases with decreasing wing core and skin densities. Reduction in material density leads to reduction in weight, which permits the use of a larger sail while still satisfying the roll stability constraint.

A different type of analysis leverages the sensitivity information directly encoded in the dual variables returned by the GP. These dual variables describe the log-derivative of the cost function with respect to each constraint and decision variable, in the locality of the optimal solution. In other words, they reveal which decision variables drive the cost function up or down, and by how much; and also, by how much the cost function would decrease if each constraint were relaxed by a given factor. For example, the most sensitive constraint in the wingsail GP—by a wide margin—is the posynomial pressure drag relationship, which was fit to offline simulation data generated using *XFOIL*. If this constraint were relaxed by 1%, the cost function would

TABLE I
MULTIOBJECTIVE OPTIMAL WINGSAIL SOLUTION ($A_W = 4.5$)

Decision Variable	Value	Units	Description
m_{tot}	2.98	[kg]	wingsail system mass
ω_n	2.17	[rad/s]	wingsail system natural frequency
U	5.86	[m/s]	velocity at wing VCE
K_g	11.91	[N·m]	wing static roll moment
K_a	2.84	[N·m]	wing aerodynamic moment
K_h	14.75	[N·m]	hull righting moment
Re_W	87,650	[]	wing Reynolds number
A_W	4.5	[]	wing aspect ratio
S_W	0.222	[m²]	wing planform area
b_W	1.000	[m]	wing span
$\bar{c}_W$	0.222	[m]	wing mean geometric chord
c_{r_W}	0.333	[m]	wing root chord
c_{t_W}	0.111	[m]	wing tip chord
λ_W	0.334	[]	wing taper ratio
τ_W	0.133	[]	wing thickness ratio
L_W	3.92	[N]	wing lift force
D_W	0.325	[N]	wing drag force
Re_T	63,710	[]	tail Reynolds number
A_T	2.25	[]	tail aspect ratio
S_T	0.0587	[m²]	tail planform area
b_T	0.363	[m]	tail span
$\bar{c}_T$	0.162	[m]	tail mean geometric chord
c_{r_T}	0.215	[m]	tail root chord
c_{t_T}	0.108	[m]	tail tip chord
λ_T	0.5	[]	tail taper ratio
τ_T	0.095	[]	tail thickness ratio
L_T	0.426	[N]	tail lift force
D_T	0.067	[N]	tail drag force
x_T	0.594	[m]	tail CG distance behind mast axis
d_T	0.555	[m]	tail AC to wing AC distance
V_T	0.66	[]	aerodynamic tail volume ratio
x_B	0.13	[m]	nose ballast distance in front of mast axis
m_B	1.655	[kg]	nose ballast mass

TABLE II
WINGSAIL GP FIXED PARAMETERS ($A_W = 4.5$)

Decision Variable	Value	Units	Description
A_W	4.5	[]	wing aspect ratio
U_{10}	8	[m/s]	air speed @ z=10m
ρ	1.225	[kg/m³]	air density
e	0.96	[]	Oswald efficiency factor
α_{max}	$\pi/4$	[rad]	wing max AOA (45 deg)
z_0	$1.545e^{-4}$	[m]	roughness length (atm BL)
z_r	0.3	[m]	root chord height above WL
x_{pin}	0.2	[]	chord-relative mast position
g	9.81	[m/s²]	gravitational acceleration
K_A	0.6851	[]	NACA profile area coefficient
ρ_c	35	[kg/m³]	wing core density
ρ_s	1.12	[kg/m²]	wing skin area density
ϕ	90	[deg]	roll stability evaluation angle

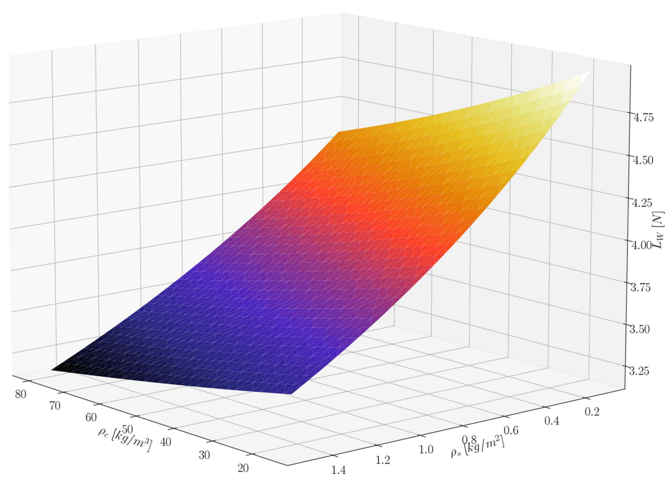

Fig. 19. Influence of wing core density (ρ_c) and skin density (ρ_s) on wing lift (L_W).

decrease by 27%. This high sensitivity conveys the importance of developing a trustworthy, high-fidelity profile drag model; moreover, it hints at the notion that the development of specialized airfoil geometries—or flapped wingsail arrangements—may result in substantial performance gains. Future refinement of the proposed approach should incorporate the effect of trailing edge flap size and deflection on wingsail performance.

The sensitivity of free and fixed variables also provide actionable insights. For example, a 1% increase in the Oswald efficiency factor yields a 0.25% decrease in the cost function, underscoring the value of developing a wing with an elliptical lift distribution; conversely, a 1% increase in the maximum angle of attack constraint incurs a 0.14% *increase* in the cost function, revealing the quantitative tension between upwind and downwind sailing requirements. When building complex GPs composed of many models and sub-models, it can be illuminating to represent sensitivity graphically, as in Fig. 20.

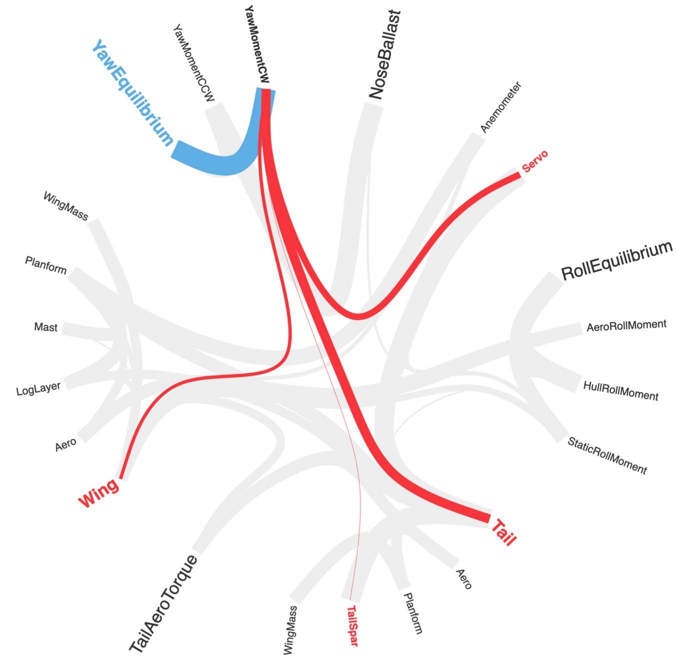

Fig. 20. Wingsail GP model diagram. Red lines indicate inherited models, blue lines indicate parent models, and the thickness of each line corresponds to the sensitivity of the constraints contained in the subsumed model.

The value of computing separate Reynolds numbers that pertain specifically to the wing and tail, at their design operating condition, cannot be understated: aerodynamic performance varies dramatically with Reynolds number, and inflation of the latter leads invariably to overestimation of former. Most small uncrewed sailboats operate in the so-called *low Reynolds number regime*; however, a great deal of confusion stems from this nebulous designation, even within the field of aerospace engineering. Typically, this term is used to refer to any flow for which the Reynolds number is less than one-million; however, while the designer of a large commercial passenger aircraft might consider a chord Reynolds number of one-million to be low, the designer of a small uncrewed aerial vehicle would almost certainly disagree. Erroneous assumptions regarding the types of airfoils which perform well at "low" Reynolds numbers have real consequences. Making reference to Fig. 8, it is apparent that the lift coefficients attainable at $Re = 200e^3$ far exceed those at $Re = 20e^3$, sometimes by more than 80%. The drag disparity is even more glaring: for most angles of attack, the drag coefficient at $Re = 20e^3$ is at least than 2.6 times larger than it would be at $Re = 200e^3$. For both lift and drag, the low-Reynolds number penalty becomes progressively more severe with increasing airfoil thickness ratio.

This Reynolds number discrepancy may explain why existing wingsails have historically tended to bias in favor of thicker airfoil sections; indeed, in the upper part of the low-Reynolds number regime ($Re \in [200e^3 : 1e^6]$), thicker airfoils can outperform thinner ones, and also exhibit more forgiving stall characteristics [51]. However, at the lower end of the low-Reynolds number regime ($Re < 100e^3$) this is no longer the case, due to the onset of laminar bubble growth and separation. Knowledge

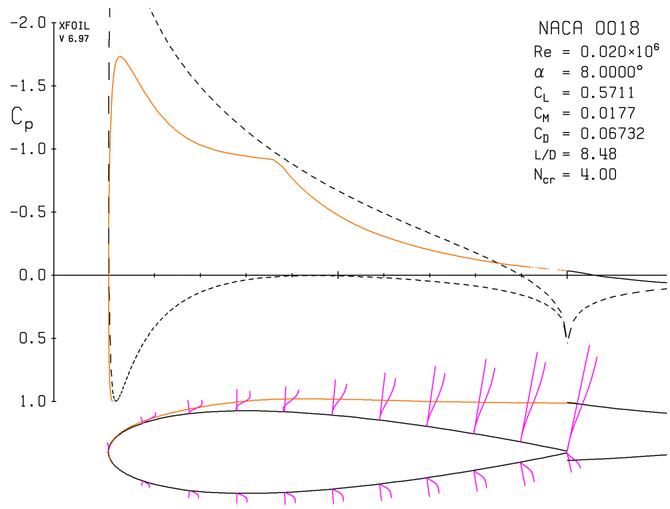

Fig. 21. Pressure distribution (upper) and boundary layer evolution (lower) of a NACA0018 airfoil operating in the very-low-Reynolds number regime ($Re = 20e^3$), simulated using *XFOIL*. Despite a relatively small angle of attack ($\alpha = 8°$), the laminar boundary layer detaches almost immediately. Boundary layer separation alters the effective shape of the airfoil, and produces a slightly positive moment coefficient.

of this phenomenon is not new: in 1985, Mueller and Reshotko [52] published a technical report for the North Atlantic Treaty Organization which remarked, "Laminar separation bubbles occur on the upper surface of most airfoils at low Reynolds numbers. These bubbles become larger as the Reynolds number decreases, usually resulting in a rapid deterioration in performance, i.e., substantial decreases in (L/D)." In brief, selection of a thicker airfoil profile in pursuit of higher efficiency and gentler stall characteristics only makes sense if the Reynolds number is large enough to prevent boundary layer separation at low angles of attack.

This contention is corroborated by Fig. 21, which displays the pressure distribution and boundary layer evolution of a NACA0018 airfoil operating in the very-low Reynolds number regime,[10] simulated using *XFOIL*. Even though the angle of attack is relatively small ($\alpha = 8°$), the laminar boundary layer separates almost immediately, before undergoing turbulent transition and nearly reattaching just aft of the mid-chord. The existence of a laminar separation bubble is evinced by the reverse flow region between the 20% and 40% chord. Thicker airfoils operating in the very-low Reynolds number regime often exhibit these features. The NACA0018 is an extremely popular airfoil profile that has been implemented on numerous rigid wingsails, despite its poor performance in the very-low Reynolds number regime.

A great deal of effort has been expended in pursuit of airfoil shapes that mitigate the deleterious effects of laminar bubble separation. Indeed, while the existence of laminar bubbles has been recognized for decades, their behavior remains somewhat poorly understood. Selig et al. [53] conducted extensive wind-tunnel analyses of various profiles, with the intent of clarifying the characteristics which correlate with better performance in the very-low Reynolds number regime. The average thickness ratio of the 54 airfoil profiles investigated was $\bar{\tau} = 0.104 \pm 0.016$ – far thinner than the airfoil profiles that are typically implemented on wingsails, which range from $\tau = 0.16$ to $\tau = 0.20$. Conversely, the wing and tail thickness ratios determined via the proposed GP optimization methodology ($\tau_W = 0.13$, $\tau_T = 0.095$) are much more similar to those which have been found empirically to provide better performance in the very-low Reynolds number regime.

The proposed methodology is not without its limitations. For one, it currently has no direct means of prioritizing natural frequency, nor of penalizing aerodynamic stall events that result from inertial overshoot, rapid changes in wind direction, or some combination thereof. Frequent stalling decimates aerodynamic performance, and yet stall tendency is rarely given much consideration at the design stage of wingsail development. In the proposed approach, domain-specific understanding is indirectly encoded in the mast position and tail separation constraints, which are defined in a way that prioritizes angular stability and natural frequency. Nevertheless, these are proxy metrics, and provide no real quantitative indication of the degree to which stall will be reduced or avoided in practice. It is possible that this shortcoming could be ameliorated by formulating the objective function as a weighted sum of aerodynamic efficiency and inverse natural frequency (or moment of inertia). However, this approach replaces one problem with another: a composite objective function of this nature is not physically meaningful, as it combines multiple quantities with different units and orders of magnitude, and there is no obvious means of determining the weights appropriately.

Along similar lines, the aerodynamic models employed in the proposed GP formulation categorically assume steady-state flow conditions, which seldom occur in the marine environment. Reliance on these simplified models may result in distorted design guidance, and overestimation of performance. Further research in this area is currently underway. The goal of this ongoing work is to clarify the effect of unsteadiness on sail performance, and assess the predictive utility of classical aerodynamic models when applied to aerodynamically-actuated wingsails.

Another potential improvement would involve the addition of a structural constraint, which relates aspect ratio to surface density. Longer, thinner wings generally require a heavier glassing schedule, or other specialized stiffening techniques. This relationship naturally lends itself to a mixed-integer description, as fiberglass and carbon fiber sheets are not available in continuously variable weights; however, mixed-integer optimization problems are significantly more difficult to solve than convex programs. A structural feasibility constraint would also be useful for larger vessels that are not so severely limited by righting moment; indeed, in the edge-case of infinite roll stability (i.e. a fixed platform), constraints would be required to ensure structural integrity of the wingsail.

One of the more meaningful improvements to the proposed methodology would be to enable some degree of efficiency

[10]Borrowing from the convention used to designate radio frequency bands, the term "very-low-Reynolds number" will be used throughout the remainder of this manuscript to describe flows with chord Reynolds number between 10 000 and 100 000.

matching between the wingsail and the hull; indeed, a highly efficient wingsail is of little use when deployed on a sluggish hull. Inspection of Fig. 2 reveals that that achieving a narrow pointing angle requires both a high lift-to-drag ratio wingsail, *and* a low-drag hull capable of converting a large portion of the aerodynamic resultant force into thrust. For slow hulls with large righting moments, it is likely beneficial to prioritize total lift, rather than lift-over-drag.

Finally, it should be noted that the proposed optimization procedure seeks to maximize upwind sailing performance at a single wind speed. For uncrewed vehicles, it may be more useful to optimize over a wide range of operating conditions, or use a different performance criterion altogether. Nevertheless, the proposed GP formulation reveals the utility of the GP approach to wingsail design. An SP extension of the proposed methodology may facilitate incorporation of important polynomial and trigonometric equality relationships inherent to wind-powered vehicle design. This more general approach should be investigated, even at the expense of global optimality guarantees.

VII. Conclusion

This manuscript details a design optimization methodology for aerodynamically-actuated wingsails. GP optimization permits the simultaneous solution of coupled decision variables, which would otherwise require iterative treatment. Although GP optimization guarantees global optimally, it does so with respect to the simplified objective function and constraints; thus, the solutions it yields may not be optimal with respect to the true objective. Nevertheless, the proposed approach provides rigorous physics-based design guidance that may reduce or eliminate entirely the need for iterative prototyping and testing cycles. A common misconception regarding the moment coefficient of symmetric airfoils is discussed in the context of mast positioning, and related to the wingsail's dynamic stability. A novel formulation of the optimal taper ratio for wings subject to a logarithmic spanwise velocity profile is also presented. The proposed GP optimization approach enables basic multiobjective optimization via design sweeps, which reveal the sensitivity of the objective function to specified parameters. Existing wingsail design methodologies rely heavily upon apocryphal heuristics, a priori assumptions, and trial-and-error prototyping. Formulating the wingsail design problem as a GP enables high-fidelity performance predictions, sensitivity analyses, and physics-informed design guidance that was not previously attainable at the conceptual design stage.

Appendix A
Additional Models

The aspect ratio of a generic wing is related to the span and planform area by the following monomial equality:

$$b = \sqrt{SA}. \tag{85}$$

The mean geometric chord of a generic wing is related to the span and planform area by the following monomial equality:

$$\bar{c} = \sqrt{\frac{S}{A}}. \tag{86}$$

The mean aerodynamic chord of a generic single-taper wing is related to the mean geometric chord and taper ratio by the following monomial equality and surrogate posynomial inequalities:

$$\bar{c}_{\text{MAC}} = \frac{4}{3}\bar{c}\nu_1(\lambda) \tag{87}$$

$$\nu_1(\lambda) \geq \frac{\lambda^2+\lambda+1}{(\lambda+1)^2} \tag{88}$$

$$\nu_1^{3.44} \geq 0.153p^{0.585} + 0.847p^{-2.15} \tag{89}$$

$$p \geq 1+2\lambda. \tag{90}$$

The constraint in p is a simple substitution that shifts lambda to positive range that does not include zero, in order to ensure λ remains bounded under log transformation. This technique is employed in all constraints involving λ.

The vertical center of gravity of a generic single-taper wing is related to the span and taper ratio by the following monomial equality and surrogate posynomial inequalities

$$\text{VCG} = \frac{\sqrt{SA}}{4}\nu_2(\lambda) \tag{91}$$

$$\nu_2(\lambda) \geq \frac{3\lambda^2+2\lambda+1}{\lambda^2+\lambda+1} \tag{92}$$

$$\nu_2 = 0.988p^{0.658} \tag{93}$$

$$p \geq 1+2\lambda. \tag{94}$$

The vertical center of effort of a generic single-taper wing in uniform flow is related to the span and taper ratio by the following monomial equality and surrogate posynomial inequalities

$$\text{VCE} = \frac{b}{3}\nu_3(\lambda) \tag{95}$$

$$\nu_3(\lambda) \geq \frac{2\lambda+1}{\lambda+1} \tag{96}$$

$$\nu_3 = 1.024p^{0.368} \tag{97}$$

$$p \geq 1+2\lambda. \tag{98}$$

The vertical center of effort of a wing that exhibits an ideal elliptical lift distribution is simply the vertical centroid of an ellipse, which is given by the following monomial equality:

$$\text{VCE}^* = \frac{4b}{3\pi} \tag{99}$$

If desired, it is possible to describe air viscosity as a function of air temperature using a monomial approximation of the Sutherland viscosity model, in the same manner as Kirschen [25]. The influence of viscosity is surprisingly nonnegligible; indeed, the GP solution is as sensitive to the Sutherland coefficient (C_μ) as it is to the reference wind speed and Oswald efficiency factor (albeit, in the opposite direction)

$$\mu = \frac{C_\mu T^{\frac{3}{2}}}{6.64T^{0.72}}. \tag{100}$$

Appendix B
Hull Photograph

Fig. 22. Photograph of the Santa Barbara class hull for which a wingsail was designed using GP optimization.

Acknowledgment

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